Building and evaluation of a PBPK model for Quinidine in adults

Version	**evaluation-OSP12.2**
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Quinidine-Model/releases/tag/vevaluation
OSP Version	12.2
Qualification Framework Version	3.5

This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

Table of Contents

- 1 Introduction
- 2 Methods
  - 2.1 Modeling strategy
  - 2.2 Data used
  - 2.3 Model parameters and assumptions
- 3 Results and Discussion
  - 3.1 Quinidine final input parameters
  - 3.2 Diagnostic plots
    - 3.2.1 Quinidine goodness-of-fit diagnostics
    - 3.2.2 3-Hydroxyquinidine goodness-of-fit diagnostics
  - 3.3 Concentration-time profiles
    - 3.3.1 Model Building
    - 3.3.2 Model Verification
- 4 Conclusion
- 5 References

1 Introduction

Quinidine is an antiarrhythmic drug. It is metabolized by CYP3A4 to 3-hydroxyquinidine, transported by P-gp (ABCB1), and inhibits CYP2D6.

This quinidine model is intended to describe plasma concentration-time profiles of quinidine and 3-hydroxyquinidine after intravenous and oral administration.

The whole-body PBPK model of quinidine was developed by Feick 2023 using published clinical data for intravenous dosing, oral dosing, repeated dosing, different oral formulations, metabolite observations, and unbound quinidine observations where available. The model was later reused in the CYP2D6 network by Rüdesheim 2025.

The presented model includes the following features:

- quinidine and 3-hydroxyquinidine as parent and metabolite compounds,
- CYP3A4-mediated formation of 3-hydroxyquinidine,
- P-gp transport and renal P-gp secretion,
- CYP2D6 inhibition by quinidine and 3-hydroxyquinidine,
- passive renal filtration and enterohepatic circulation,
- intravenous infusion, oral solution, capsule, tablet, and multiple-dose applications.

2 Methods

2.1 Modeling strategy

The general concept of building a PBPK model has previously been described by Kuepfer et al. (Kuepfer 2016). Relevant information on anthropometric and physiological parameters in adults was gathered from the literature and incorporated into PK-Sim as default values for adult simulations (Willmann 2007).

The applied activity and variability of plasma proteins and active processes integrated into PK-Sim are described in the publicly available PK-Sim Ontogeny Database or otherwise referenced for the specific process.

The quinidine model was developed as a parent-metabolite PBPK model for quinidine and 3-hydroxyquinidine by Feick 2023. Model development used intravenous and oral clinical data to inform distribution, oral absorption, P-gp transport, CYP3A4-mediated metabolism, metabolite disposition, and renal elimination.

Clinical studies used for model building were selected to cover key structural information, including intravenous dosing, nonlinear oral exposure, repeated dosing, and metabolite observations. Verification simulations used independent oral and intravenous study arms, different dose levels and formulations, salt-intake conditions, and quinidine-only control arms from coadministration studies. This split preserves the training and test logic used for model evaluation.

The intravenous data support systemic distribution, renal elimination, and parent-metabolite disposition without the need to estimate oral absorption. Oral and repeated-dose studies then support absorption, nonlinear exposure, enterohepatic circulation, and transporter-related assumptions. The use of parent, metabolite, and unbound quinidine observations provides complementary checks on the structural model.

The model contains CYP3A4 metabolism to 3-hydroxyquinidine, P-gp transport, P-gp-mediated renal secretion, passive renal filtration, enterohepatic circulation, and CYP2D6 inhibition. Parent quinidine, 3-hydroxyquinidine, and unbound quinidine observations are evaluated separately where available.

The evaluated applications include intravenous infusion, oral solution, oral capsule, oral tablet, single-dose, and multiple-dose dosing. The major proteins and processes represented explicitly are CYP3A4, P-gp, CYP2D6 inhibition, plasma protein binding, passive renal filtration, renal secretion, and enterohepatic circulation. These processes make the quinidine model broader than a clearance-only substrate model.

The report evaluates total quinidine, unbound quinidine, and 3-hydroxyquinidine profiles where available. Model-building profiles are used to establish the structural assumptions, while verification profiles test independent doses, formulations, and quinidine-only control arms without additional fitting.

Details about input data are provided in Section 2.2. Details about the structural model and assumptions are provided in Section 2.3.

2.2 Data used

In vitro and physicochemical data

Drug-dependent parameters for quinidine and 3-hydroxyquinidine were taken from the published PBPK model and its supplement (Feick 2023).

Parameter	Unit	Value	Source	Description
Quinidine				
MW	g/mol	324.42	Literature (DrugBank 2006, DrugBank Quinidine)	Molecular weight.
pK_a,base	-	4.02	Literature (Takano 2016)	pKa of a basic ionization site.
pK_a,base	-	9.05	Literature (DrugBank 2006)	pKa of a basic ionization site.
pK_a,acid	-	13.89	Literature (DrugBank Quinidine)	pKa of an acidic ionization site.
Solubility	g/L	11.11	Literature (Grube 2009)	Aqueous solubility of quinidine sulfate at pH 7.0.
logP	-	2.51	Literature (DrugBank 2006, DrugBank Quinidine)	Octanol-water partition coefficient of the neutral species.
f_u	%	21	Calculated (Watanabe 2018)	Fraction unbound in plasma.
$K_{m,P\text{-}gp}$	µmol/L	0.23	Literature (Tachibana 2010)	Michaelis constant for quinidine.
$k_{cat,P\text{-}gp}$	1/min	0.77	Optimized (Feick 2023, Table S5)	Catalytic rate constant for quinidine.
$K_{i,P\text{-}gp}$	µmol/L	0.10	Literature (Lumen 2010)	Inhibition constant for P-gp.
$K_{m,CYP3A4}$	µmol/L	51.8	Literature (Nielsen 1999, Austin 2002)	Michaelis constant for formation of 3-hydroxyquinidine from quinidine.
$k_{cat,CYP3A4}$	1/min	2.21	Optimized (Feick 2023, Table S5)	Catalytic rate constant for formation of 3-hydroxyquinidine from quinidine.
$K_{m,CYP3A4}$	µmol/L	65.03	Literature (Nielsen 1999, Austin 2002)	Michaelis constant for formation of other metabolites from quinidine.
$k_{cat,CYP3A4}$	1/min	3.84	Optimized (Feick 2023, Table S5)	Catalytic rate constant for formation of other metabolites from quinidine.
$K_{i,CYP2D6}$	µmol/L	0.017	Literature, estimated in vivo value (Moghadamnia 2003)	Inhibition constant for CYP2D6.
GFR fraction	-	1.00	Assumed (Feick 2023, Table S5)	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1.00	Assumed (Feick 2023, Table S5)	Fraction released continuously into bile.

Parameter	Unit	Value	Source	Description
Partition coefficients	-	Berezhkovskiy	Calculated (Berezhkovskiy 2004)	Tissue-to-plasma partition coefficients calculated with the Berezhkovskiy method.
Cellular permeabilities	cm/min	7.99×10^{-3}	Calculated, PK-Sim Standard (Feick 2023, Table S5)	Calculated cellular permeability.
P_{int}	cm/min	6.47×10^{-6}	Optimized (Feick 2023, Table S5)	Specific transcellular intestinal permeability.
t_{50}	min	8.76	Calculated from literature (Nguyen 2006, Zhang 2010)	Time to dissolve 50% of the dose.
Weibull shape	-	0.42	Calculated from literature (Nguyen 2006, Zhang 2010)	Shape parameter of the Weibull dissolution function.
t_{50}, sustained release	min	142	Optimized	Time to dissolve 50% of the dose. Used only for Kaukonen 1997.
Weibull shape, sustained release	-	0.84	Optimized	Shape parameter of the Weibull dissolution function. Used only for Kaukonen 1997.
3-Hydroxyquinidine				
MW	g/mol	340.42	Literature (DrugBank 3-Hydroxyquinidine)	Molecular weight.
pK_a,base	-	4.03	Literature (Chemicalize 2020)	pKa of a basic ionization site.
pK_a,base	-	8.63	Literature (DrugBank 2006, DrugBank 3-Hydroxyquinidine)	pKa of a basic ionization site.
pK_a,acid	-	13.55	Literature (DrugBank 2006, DrugBank 3-Hydroxyquinidine)	pKa of an acidic ionization site.
Solubility	g/L	12.57	Literature (Chemicalize 2020)	Aqueous solubility at pH 7.4.
logP	-	1.66	Literature (Chemicalize 2020)	Octanol-water partition coefficient of the neutral species.
f_u	%	31	Calculated (Watanabe 2018)	Fraction unbound in plasma.

Parameter	Unit	Value	Source	Description
CL_{CYP3A4}	1/min	0.08	Optimized (Feick 2023, Table S5)	CYP3A4-mediated clearance.
CL_{hep}	1/min	0.45	Optimized (Feick 2023, Table S5)	Unspecific hepatic clearance.
$K_{i,CYP2D6}$	µmol/L	2.30	Literature (Ching 1995)	Inhibition constant for CYP2D6.
GFR fraction	-	1.00	Assumed (Feick 2023, Table S5)	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1.00	Assumed (Feick 2023, Table S5)	Fraction released continuously into bile.
Partition coefficients	-	Berezhkovskiy	Calculated (Berezhkovskiy 2004)	Tissue-to-plasma partition coefficients calculated with the Berezhkovskiy method.
Cellular permeabilities	cm/min	8.45×10^{-4}	Calculated, PK-Sim Standard (Feick 2023, Table S5)	Calculated cellular permeability.
P_{int}	cm/min	2.94×10^{-6}	Calculated (Feick 2023, Table S5)	Specific transcellular intestinal permeability.

Table 1: Drug-dependent physicochemical, distribution, absorption, metabolism, transport, elimination, and inhibition parameters from Supplement Table S5 of Feick 2023. Values are displayed at the precision reported in the supplement. Optimized values are shown without their confidence intervals.

Clinical data

The model project contains 70 observed-data records. These are 53 plasma records (42 total quinidine, 2 unbound quinidine, eight 3-hydroxyquinidine, and 1 quinidine record that no simulation uses) and 17 urine records (12 quinidine and five 3-hydroxyquinidine). The evaluation uses 43 simulated study arms with quinidine plasma concentration-time profiles (Table 2). Ten profiles were used for model building and 33 profiles were used for model verification.

Source	Dose / schedule*	Age [years]	Weight [kg]	Sex	N	Form.
Andreasen 2007+	200 mg quinidine sulfate, oral	21–26	62–85	Male	10	Capsule
Bleske 1990	400 mg quinidine sulfate, oral	22–34	NR	Male	8	Tablet
Bolme 1977	200 mg quinidine sulfate three times daily, oral	26–33	73.4 (62–90)	Male	5	Tablet
Bolme 1977	300 mg quinidine sulfate three times daily, oral	26–33	73.4 (62–90)	Male	5	Tablet
Bolme 1977	400 mg quinidine sulfate three times daily, oral	26–33	68 (62–75)	Male	3	Tablet
Ching 1991	400 mg quinidine sulfate, oral	22–29	60–94	Male	8	Tablet
Damkier 1999 (fluvoxamine)	200 mg quinidine sulfate, oral	NR	NR	Male	6	Tablet
Damkier 1999 (rifampicin)	200 mg quinidine sulfate, oral	NR	NR	Male	6	Tablet
Darbar 1997	187.5 mg quinidine base, intravenous infusion over 30 min, high-salt diet	28.6 (22–37)	71.5	Male	9	Solution
Darbar 1997	187.5 mg quinidine base, intravenous infusion over 30 min, low-salt diet	28.6 (22–37)	68.4	Male	9	Solution
Darbar 1997	600 mg quinidine sulfate, oral, high-salt diet	28.6 (22–37)	71.5	Male	9	Tablet
Darbar 1997	600 mg quinidine sulfate, oral, low-salt diet	28.6 (22–37)	68.4	Male	9	Tablet
Edwards 1987	400 mg quinidine sulfate, oral	23–34	NR	Male	6	Tablet
Fremstad 1979	0.5 mmol quinidine gluconate, intravenous infusion over 60 min	NR	NR	NR	7	Solution
Fremstad 1979	1 mmol quinidine gluconate, intravenous infusion over 60 min	NR	NR	NR	6	Solution
Frigo 1977	600 mg quinidine sulfate, oral	26.4 (23–37)	67.1 (60–76)	Male	8	Tablet

Source	Dose / schedule*	Age [years]	Weight [kg]	Sex	N	Form.
Guentert 1979	3.74 mg/kg quinidine base, intravenous infusion over 25 min, subject EF first period	23	70.4	Male	1	Solution
Guentert 1979	3.74 mg/kg quinidine base, intravenous infusion over 25 min, subject EF second period	23	72.2	Male	1	Solution
Guentert 1979	3.74 mg/kg quinidine base, intravenous infusion over 25 min, subject RL	32	82	Male	1	Solution
Hardy 1988	400 mg quinidine sulfate, oral	25–38	NR	Male	6	Tablet
Kaukonen 1997	100 mg quinidine sulfate, oral	25 (21–32)	64 (41–80)	56% female	9	Tablet
Kolb 1984	400 mg quinidine sulfate, oral	21–35	NR	Male	9	Tablet
Laganière 1996	200 mg quinidine sulfate, oral	24 (19–37)	75 (65–101)	Male	12	Tablet
Maeda 2011+	0.1 mg quinidine sulfate, oral	27	NR	Male	7	Solution
Maeda 2011+	1 mg quinidine sulfate, oral	27	NR	Male	7	Solution
Maeda 2011+	10 mg quinidine sulfate, oral	27	NR	Male	7	Solution
Maeda 2011+	100 mg quinidine sulfate, oral	27	NR	Male	7	Solution
Mason 1976	200 mg quinidine sulfate, oral	22–40	NR	11% female	13	Solution
Mason 1976	200 mg quinidine sulfate, oral	22–40	NR	11% female	13	Tablet[a]
Mason 1976	200 mg quinidine sulfate, oral	22–40	NR	11% female	13	Capsule
Ochs 1978	400 mg quinidine sulfate, oral	28.9 (27–31)	68.4 (57.7–79.5)	43% female	7	Tablet
Ochs 1978+	400 mg then 200 mg quinidine sulfate four times daily, oral	28.9 (27–31)	68.4 (57.7–79.5)	43% female	7	Tablet
Ochs 1980+	300 mg quinidine base, intravenous infusion over 15 min, subject B	27	60	Male	1	Solution

Source	Dose / schedule*	Age [years]	Weight [kg]	Sex	N	Form.
Ochs 1980+	300 mg quinidine base, intravenous infusion over 15 min, subject A	23	80	Male	1	Solution
Rao 1995	250 mg quinidine sulfate, oral	18–26	48–62	Male	8	Capsule
Shin 2007	4 mg/kg quinidine base, intravenous infusion over 20 min	27.7	60.7	Female	6	Solution
Shin 2007	4 mg/kg quinidine base, intravenous infusion over 20 min	22.7	53.4	Female	12	Solution
Shin 2007	4 mg/kg quinidine base, intravenous infusion over 20 min	26.2	69.8	Male	7	Solution
Shin 2007	4 mg/kg quinidine base, intravenous infusion over 20 min	22.1	66.5	Male	12	Solution
Strum 1977+	400 mg quinidine sulfate, oral, treatment A	20–37	NR	Male	11	Immediate-release tablet
Strum 1977+	400 mg quinidine sulfate, oral, treatment B	20–37	NR	Male	11	Immediate-release tablet
Strum 1977	400 mg quinidine sulfate, oral, treatment C	20–37	NR	Male	11	Immediate-release tablet
Strum 1977	400 mg quinidine sulfate, oral, treatment D	20–37	NR	Male	11	Immediate-release tablet

Table 2: Clinical quinidine concentration-time profiles used for model building and verification. Data are from Supplement Table S4 of Feick 2023. *: Single dose unless otherwise specified. Oral doses are given as quinidine sulfate. The simulations use the quinidine base equivalent (factor 0.8287). Intravenous doses are given as quinidine base, except Fremstad 1979. The Fremstad 1979 doses were simulated as 162.2 mg and 324.4 mg quinidine base. NR: not reported. +: data used for model building. [a]: The observed data of this arm in the model project were identical to those of the solution arm at all 11 time points. They were removed. The figure shows the simulation only. The tablet data were not checked against Mason 1976.

2.3 Model parameters and assumptions

Absorption

The model includes intravenous infusion and oral quinidine applications. Intravenous simulations do not require an absorption process. Oral quinidine absorption is represented with a Weibull formulation and compound-specific intestinal permeability.

The Weibull parameters were calculated from literature dissolution data with DDSolver. `Specific intestinal permeability` was optimized to describe the available oral formulation data. The clinical dataset includes solution, capsule, tablet, and multiple-dose scenarios, so formulation assumptions are central to the oral model evaluation.

The intravenous studies constrain systemic disposition separately from oral absorption. Oral simulations then use the same systemic disposition assumptions and add the Weibull formulation parameters. This is relevant because quinidine exposure can be affected by absorption rate, P-gp transport, metabolism, renal secretion, and enterohepatic circulation.

The same formulation concept is used across oral studies, with study-specific applications defining the administered dose and dosing schedule. The model does not use separate fitted absorption parameters for each clinical study arm.

Distribution

Quinidine plasma protein binding was represented by a fraction unbound of 21%, and 3-hydroxyquinidine was represented with a fraction unbound of 31%, as summarized in Section 2.2.

Partition coefficients were calculated with the Berezhkovskiy method. Unbound quinidine profiles are included where available, which helps evaluate whether total and unbound concentrations are described consistently.

Separate compound properties are used for quinidine and 3-hydroxyquinidine. This is necessary because metabolite concentrations cannot be interpreted only as scaled parent concentrations. The availability of unbound quinidine data provides an additional check on whether the protein-binding assumption is compatible with the total-concentration profiles.

Distribution was not fitted separately by formulation, dose, or interaction scenario. The same adult physiological assumptions and compound-specific binding inputs are used across the evaluated simulations.

Metabolism and elimination

Metabolism, transport, renal elimination, and inhibition processes are represented in the model.

- CYP3A4

Quinidine is metabolized by CYP3A4 to 3-hydroxyquinidine. CYP3A4 K_m was taken from *in vitro* data corrected for microsomal binding, while CYP3A4 k_{cat} was optimized.

CYP3A4 controls parent depletion and metabolite formation. The same enzymatic pathway therefore affects quinidine and 3-hydroxyquinidine profiles in opposite directions. Metabolite data are important for evaluating whether CYP3A4 formation is plausible and not only whether parent exposure is reproduced.

- P-gp (ABCB1)

P-gp transport is implemented for quinidine. The model includes P-gp-mediated transport relevant for intestinal handling and renal secretion.

P-gp was included because quinidine is a known transporter substrate and transporter-mediated renal secretion is part of the model structure. This process can influence both oral absorption and renal elimination, so it is evaluated together with formulation and renal profiles rather than as an isolated parameter.

- CYP2D6 inhibition

Quinidine and 3-hydroxyquinidine are represented as CYP2D6 inhibitors. Their CYP2D6 K_i values are summarized in Section 2.2.

- Renal and residual elimination

The model includes passive renal filtration, renal secretion through P-gp, enterohepatic circulation, 3-hydroxyquinidine CYP3A4 clearance, and 3-hydroxyquinidine unspecific hepatic clearance.

Renal elimination combines passive filtration with active secretion. Enterohepatic circulation is included as an assumption with a continuous fraction of 1 (Feick 2023, Table S5). The metabolite unspecific hepatic clearance accounts for downstream 3-hydroxyquinidine elimination.

3 Results and Discussion

The PBPK model for quinidine was developed and evaluated with clinical pharmacokinetic data after intravenous and oral administration. The evaluation covers intravenous infusion, oral solution, capsule, and tablet applications, single-dose and multiple-dose regimens, quinidine and 3-hydroxyquinidine concentration-time profiles, and total and unbound quinidine observations where available.

The model-building data supported systemic disposition, absorption, formulation behavior, CYP3A4-mediated 3-hydroxyquinidine formation, renal elimination, P-gp transport, and enterohepatic circulation. Building studies included intravenous and oral quinidine data from Ochs 1980, Maeda 2011, Andreasen 2007, Ochs 1978, and Strum 1977. Verification used independent oral and intravenous studies, multiple formulations, metabolite observations, and quinidine-only control arms as summarized in Section 2.2.

The model includes CYP3A4-mediated formation of 3-hydroxyquinidine and other metabolites, P-gp transport, passive renal filtration, active renal secretion, enterohepatic circulation, and CYP2D6 inhibition. The 3-hydroxyquinidine observations inform metabolite formation and elimination.

The next sections show:

1. the final model input parameters for the building blocks in Section 3.1.
2. the overall goodness of fit in Section 3.2.
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and model verification in Section 3.3.

3.1 Quinidine final input parameters

The compound parameter values of the final PBPK model are illustrated below.

Compound: Quinidine

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	11.11 mg/ml	Publication-In Vitro-Grube 2009	quinidine sulfate solubility (Grube 2009)	True
Reference pH	7	Publication-In Vitro-Grube 2009	quinidine sulfate solubility (Grube 2009)	True
Lipophilicity	2.51 Log Units	Internet-Unknown-DrugBank	Measurement	True
Fraction unbound (plasma, reference value)	0.2108366055	Publication-Other-Predicted, Watanabe et al. 2018	Measurement	True
Specific intestinal permeability (transcellular)	6.472015277E-06 cm/min	Parameter Identification-Parameter Identification-Optimized	FIT	True
Is small molecule	Yes			
Molecular weight	324.42 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Berezhkovskiy
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Other-Assumption

Inhibition: CYP2D6-Moghadamnia 2003

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
Ki	0.017 µmol/l	Publication-Other-Moghadamnia 2003

Inhibition: ABCB1-Lumen 2010

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Ki	0.1 µmol/l	Publication-Lumen 2010

Metabolizing Enzyme: CYP3A4-Nielsen 1999 (3-Hydroxyquinidine), adjusted for fu,inc (70%)

Molecule: CYP3A4

Metabolite: 3-Hydroxyquinidine

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	1.097 nmol/min/mg mic. protein	Publication-In Vitro-Nielsen 1999
Km	51.8 µmol/l	Publication-In Vitro-Nielsen 1999, corrected for fu,inc (70%)
kcat	2.2098117205 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP3A4-Nielsen 1999 (sink), adjusted for fu,inc (70%)

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0.27 nmol/min/mg mic. protein	Publication-In Vitro-Nielsen 1999
Km	65.03 µmol/l	Publication-In Vitro-Nielsen 1999, corrected for fu,inc (70%)
kcat	3.8413006533 1/min	Parameter Identification-Parameter Identification-Optimized

Transport Protein: ABCB1-Tachibana 2010

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Transporter concentration	1 µmol/l	
Vmax	0 µmol/l/min	
Km	0.23 µmol/l	Publication-In Vitro-Tachibana 2010
kcat	0.7653583636 1/min	Parameter Identification-Parameter Identification-Optimized

Compound: 3-Hydroxyquinidine

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	12.57 mg/ml	Unknown-Chemicalize	Measurement	True
Reference pH	7.4	Unknown-Chemicalize	Measurement	True
Lipophilicity	1.66 Log Units	Internet-Unknown-Chemicalize	Predicted	True
Fraction unbound (plasma, reference value)	0.3097	Publication-Other-Predicted, Watanabe et al. 2018	Measurement	True
Is small molecule	Yes			
Molecular weight	340.4162 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Berezhkovskiy
Cellular permeabilities	PK-Sim Standard

Processes

Inhibition: CYP2D6-Ching 1995

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
Ki	2.3 µmol/l	Publication-Ching 1995

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Other-Assumption-Assumed

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.15	
Lipophilicity (experiment)	2.41 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	0.4523113726 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP3A4-Assumed

Species: Human

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
Intrinsic clearance	0 l/min	
Specific clearance	0.0784146784 1/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

The goodness-of-fit diagnostics are presented separately for quinidine and 3-hydroxyquinidine. This separation prevents parent and metabolite prediction errors from being hidden in a combined statistic or an ambiguous legend. The GOF diagnostics use plasma data only. Urine data are not included.

3.2.1 Quinidine goodness-of-fit diagnostics

Below you find the goodness-of-fit visual diagnostic plots for quinidine plasma concentration data used in the model evaluation. Total and unbound quinidine are shown as separate groups.

The first plot shows observed versus simulated plasma concentration, the second residuals versus time. Each residual is the natural logarithm of the observed concentration minus the natural logarithm

of the simulated concentration.

Table 3-1: GMFE for Quinidine goodness-of-fit diagnostics

Group	GMFE
Quinidine	1.35
Quinidine unbound	1.24
All	1.35

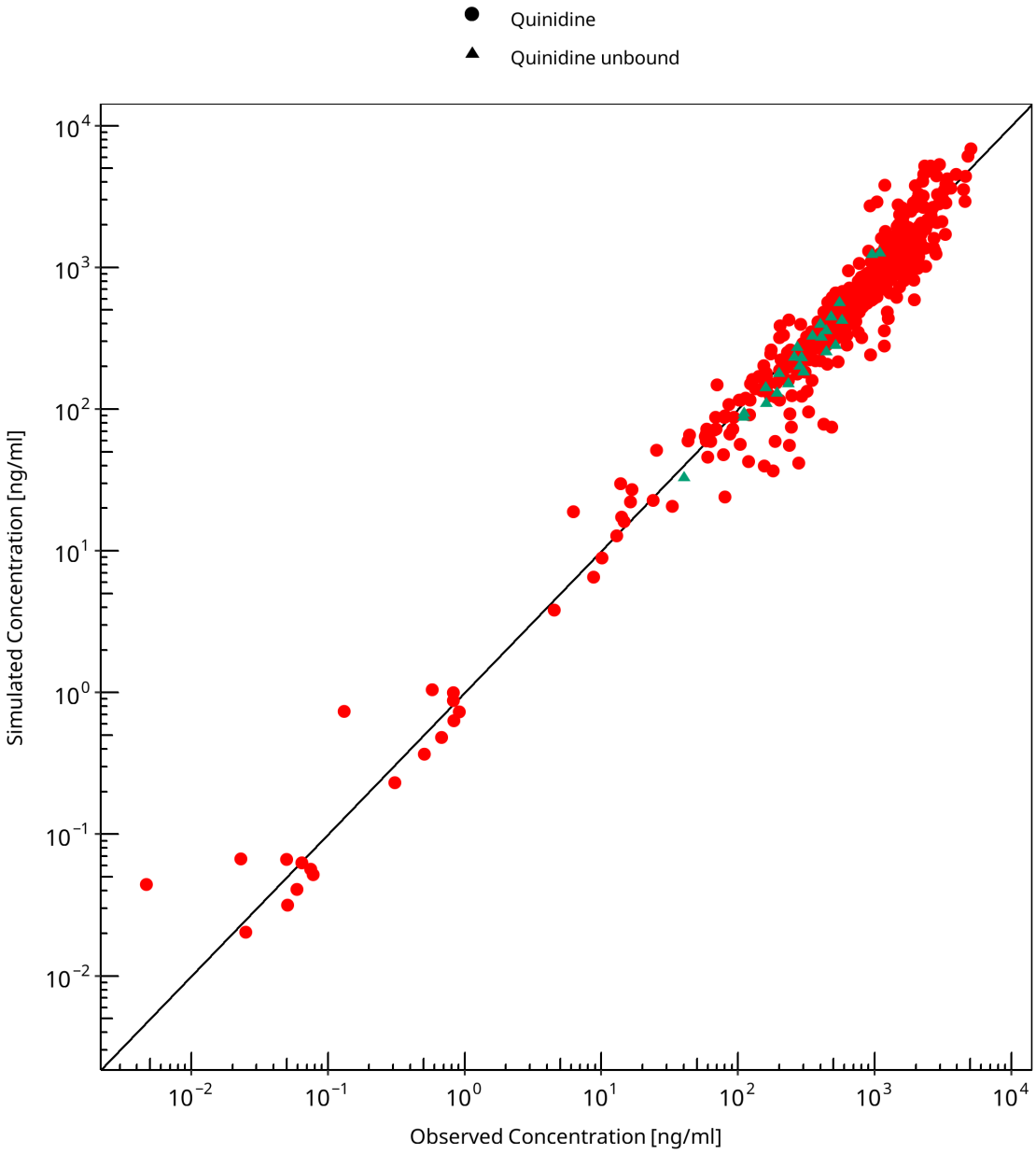


Figure 3-1: Quinidine goodness-of-fit diagnostics

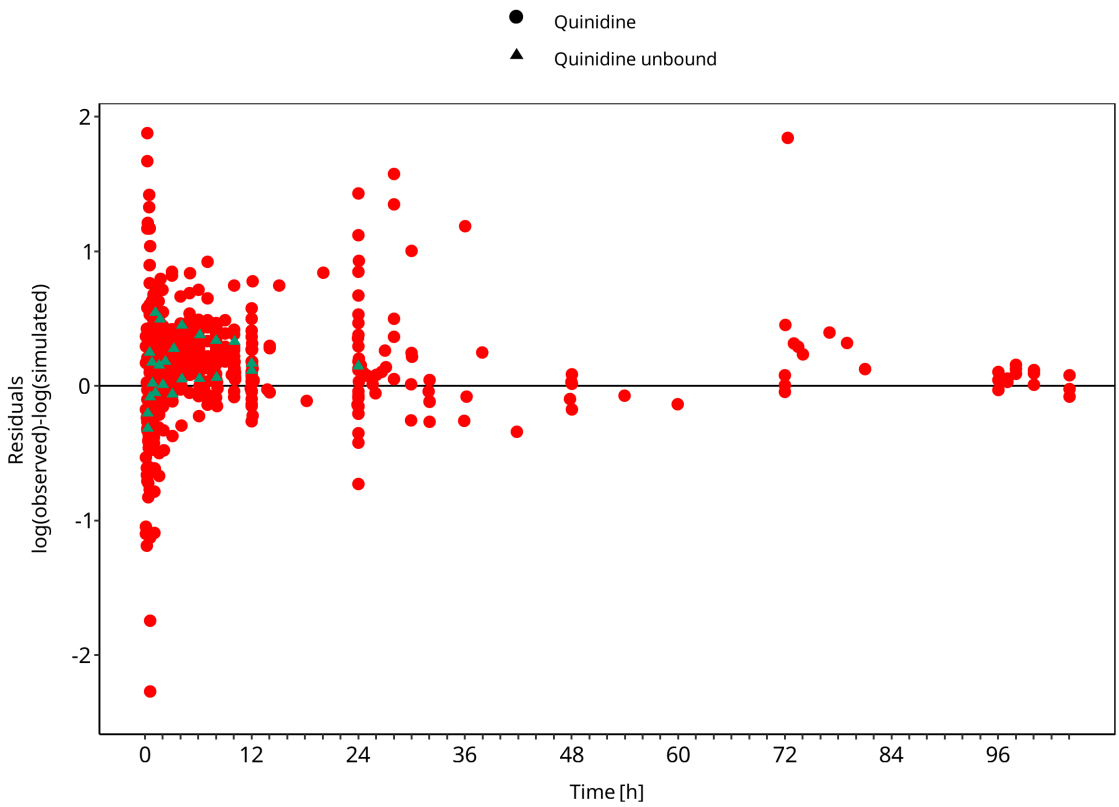


Figure 3-2: Quinidine goodness-of-fit diagnostics

3.2.2 3-Hydroxyquinidine goodness-of-fit diagnostics

Below you find the goodness-of-fit visual diagnostic plots for 3-hydroxyquinidine plasma concentration data used in the model evaluation.

The first plot shows observed versus simulated plasma concentration, the second residuals versus time. Each residual is the natural logarithm of the observed concentration minus the natural logarithm of the simulated concentration.

Table 3-2: GMFE for 3-Hydroxyquinidine goodness-of-fit diagnostics

Group	GMFE
3-Hydroxyquinidine	1.45

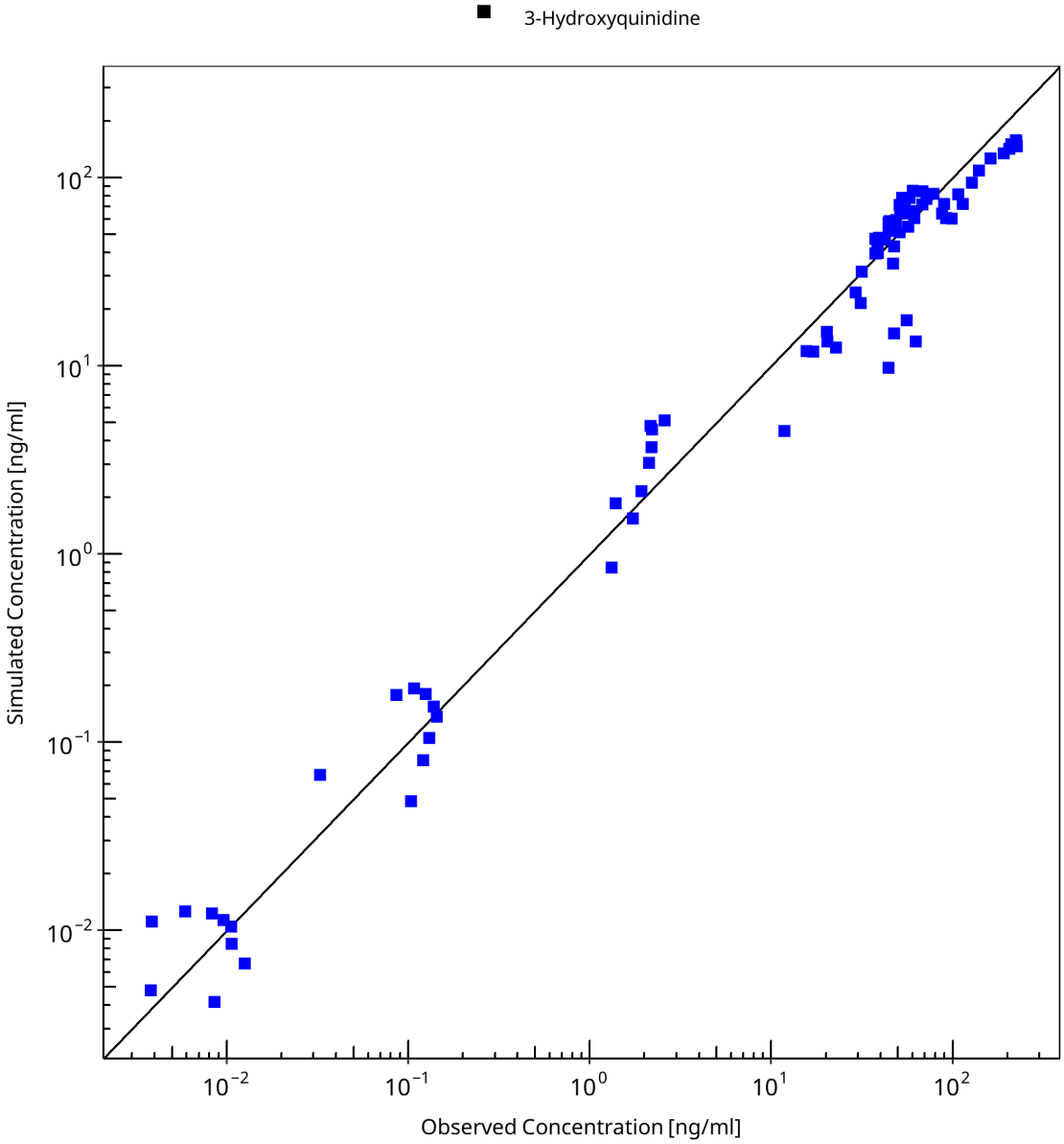


Figure 3-3: 3-Hydroxyquinidine goodness-of-fit diagnostics

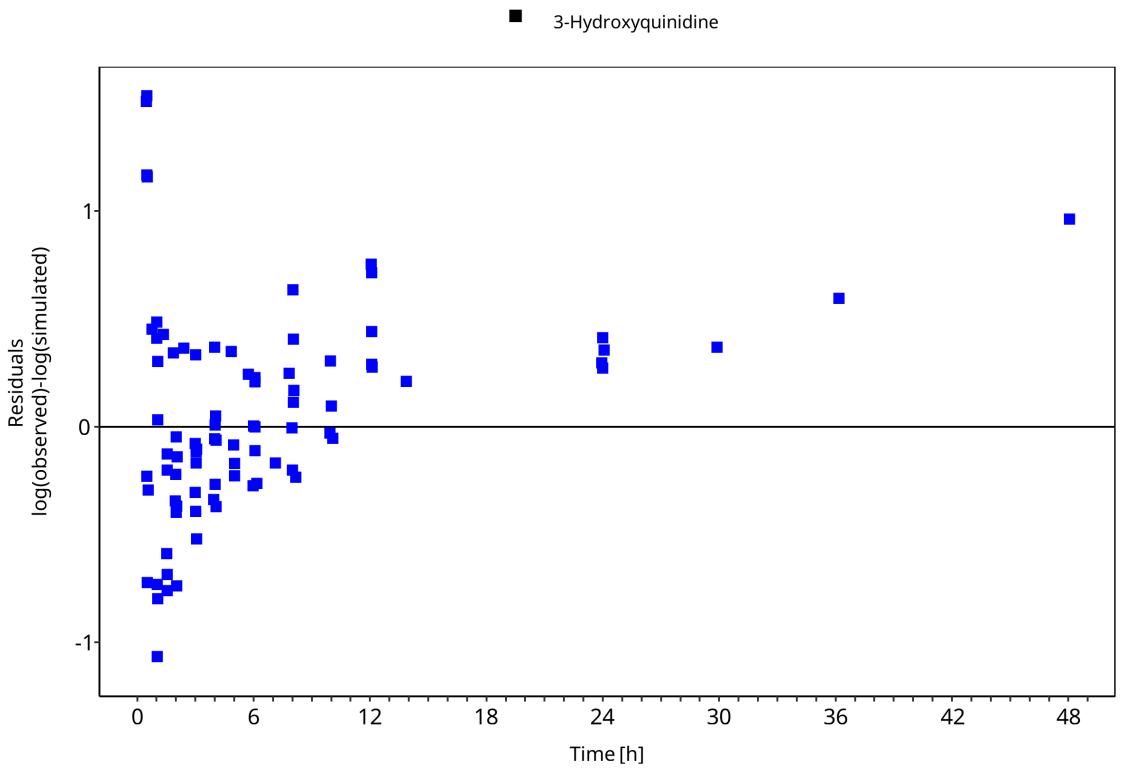


Figure 3-4: 3-Hydroxyquinidine goodness-of-fit diagnostics

3.3 Concentration-time profiles

The concentration-time profiles show simulated and observed data for the model-building and model-verification datasets listed in Section 2.2. The profiles include intravenous and oral quinidine administration, total and unbound quinidine, and 3-hydroxyquinidine where available.

3.3.1 Model Building

The following model-building profiles show the quinidine and 3-hydroxyquinidine data used to establish systemic disposition, oral input, and metabolite formation. They should be interpreted by route, formulation, and analyte.

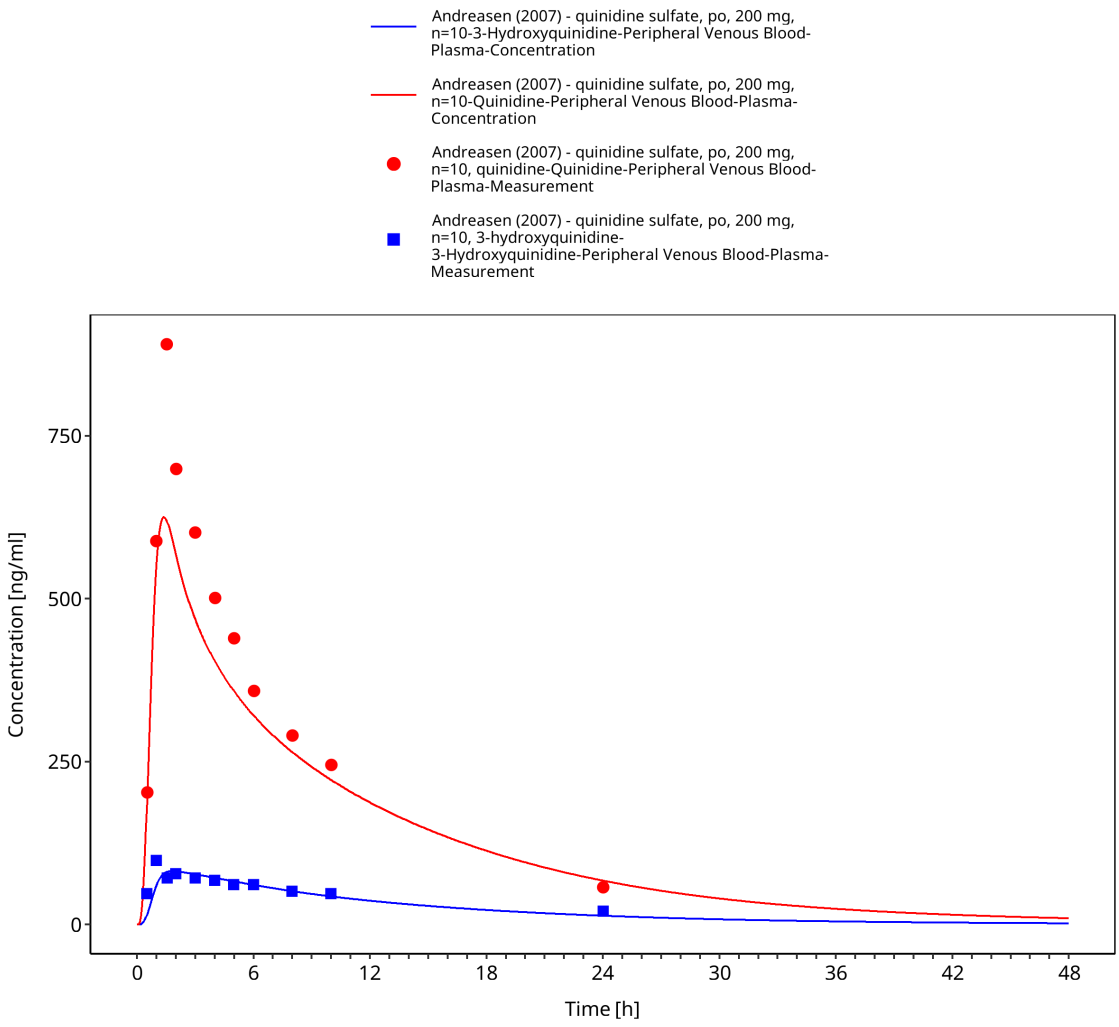


Figure 3-5: Andreasen 2007: quinidine sulfate, po, 200 mg, n=10

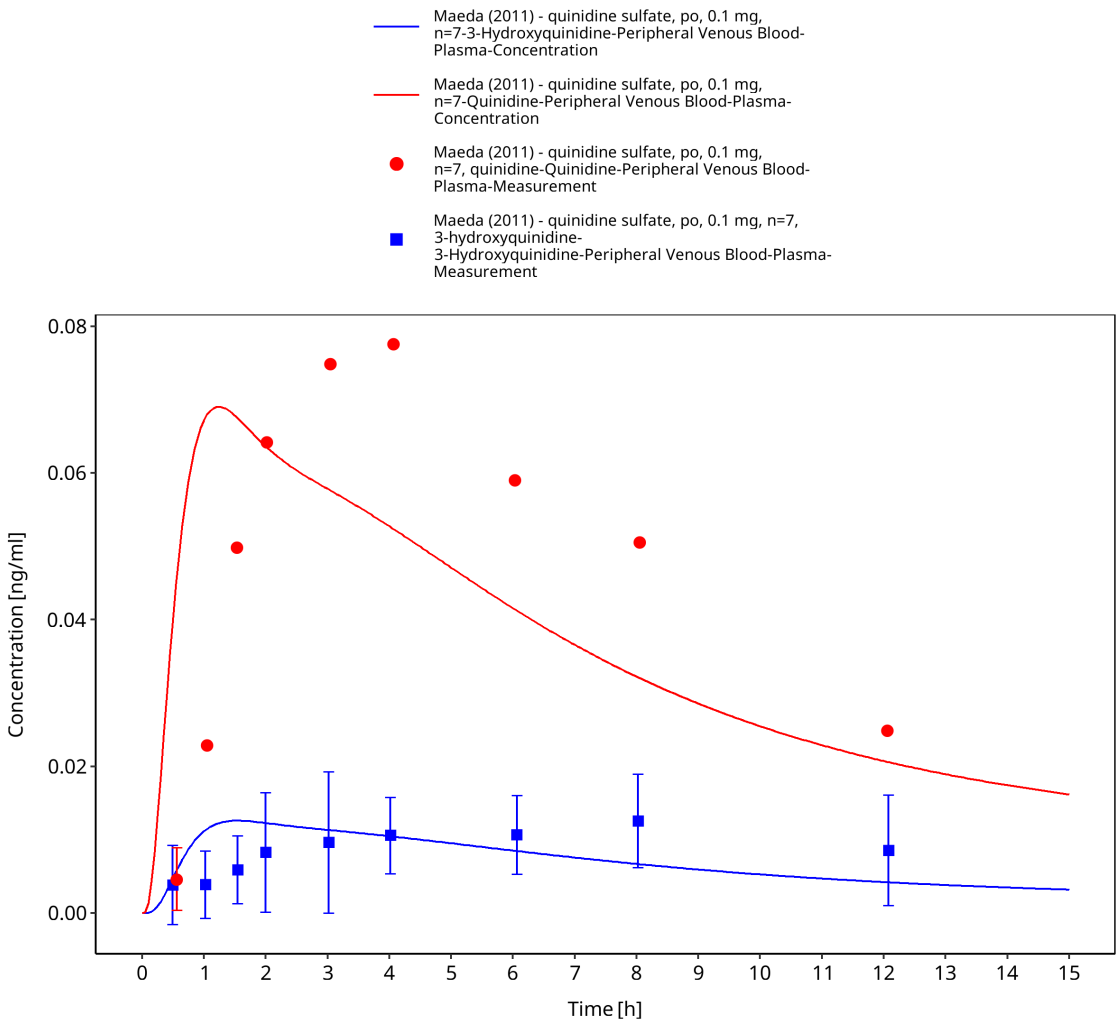


Figure 3-6: Maeda 2011: quinidine sulfate, po, 0.1 mg, n=7

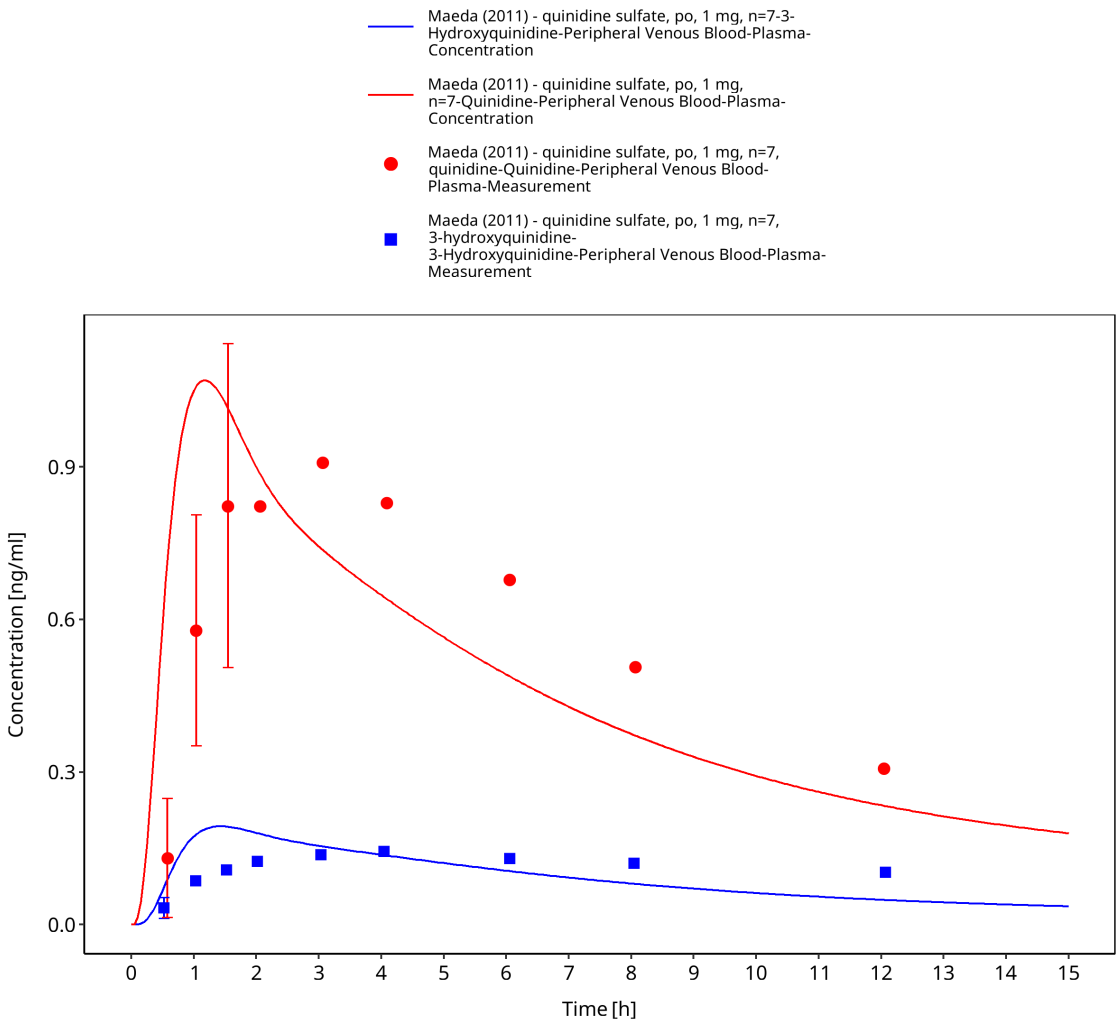


Figure 3-7: Maeda 2011: quinidine sulfate, po, 1 mg, n=7

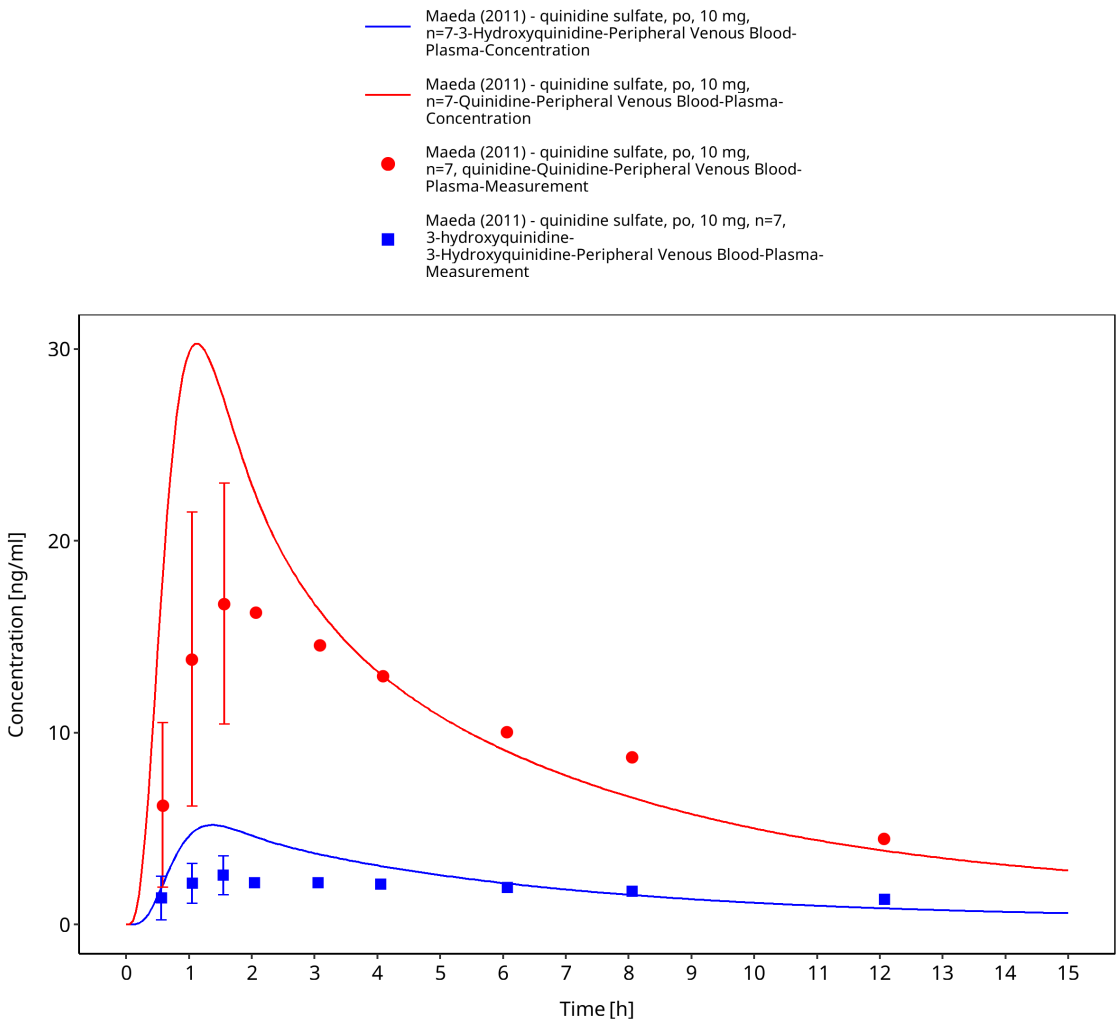


Figure 3-8: Maeda 2011: quinidine sulfate, po, 10 mg, n=7

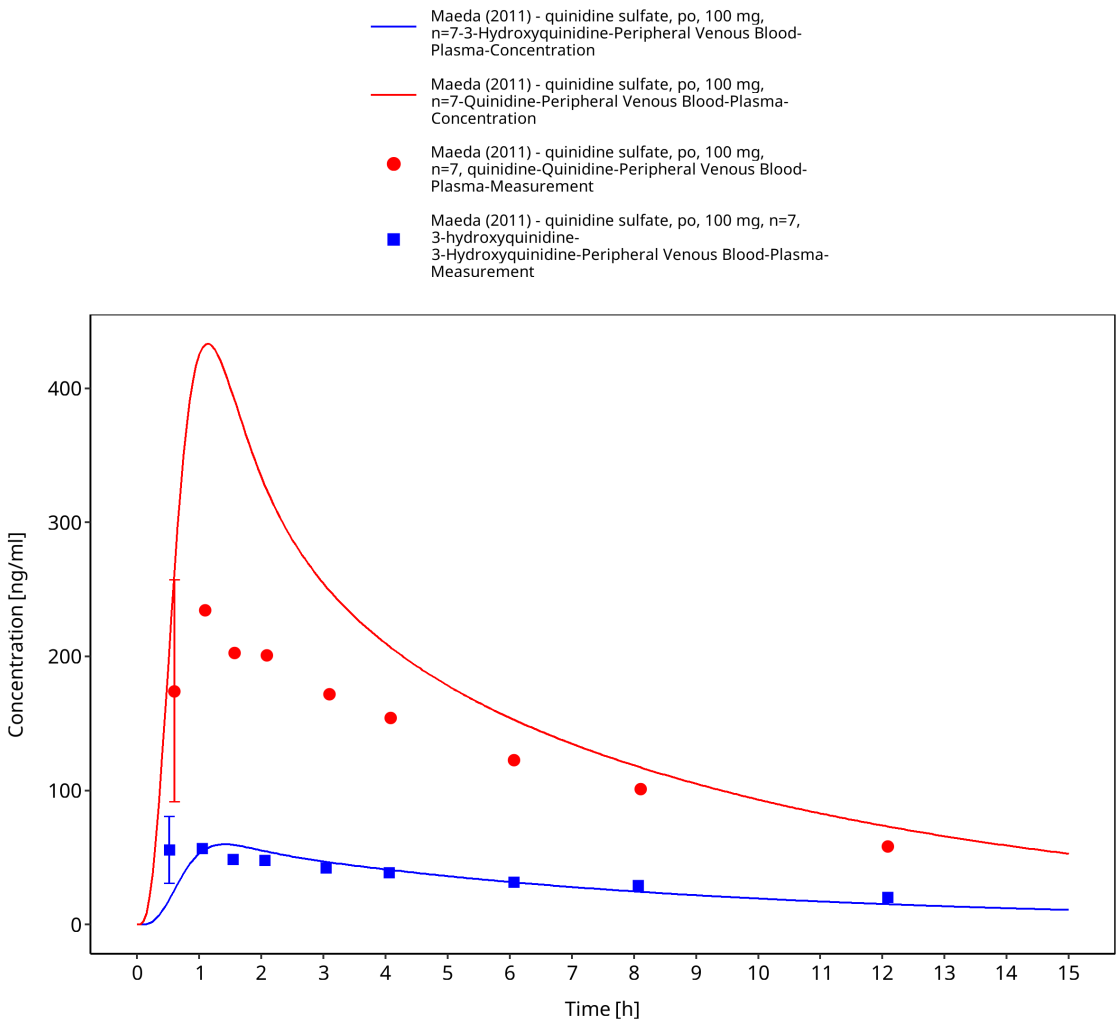


Figure 3-9: Maeda 2011: quinidine sulfate, po, 100 mg, n=7

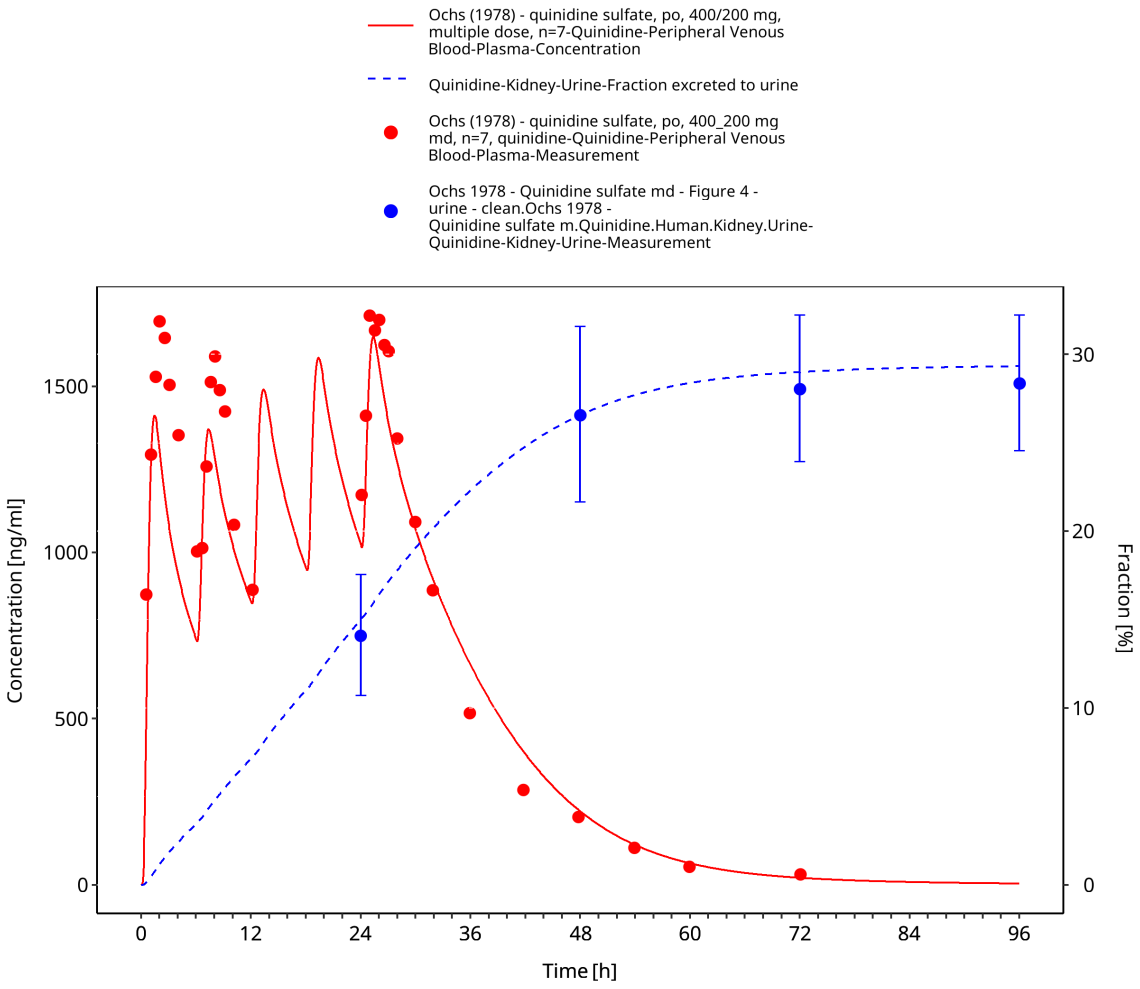


Figure 3-10: Ochs 1978: quinidine sulfate, po, 400/200 mg multiple dose, n=7

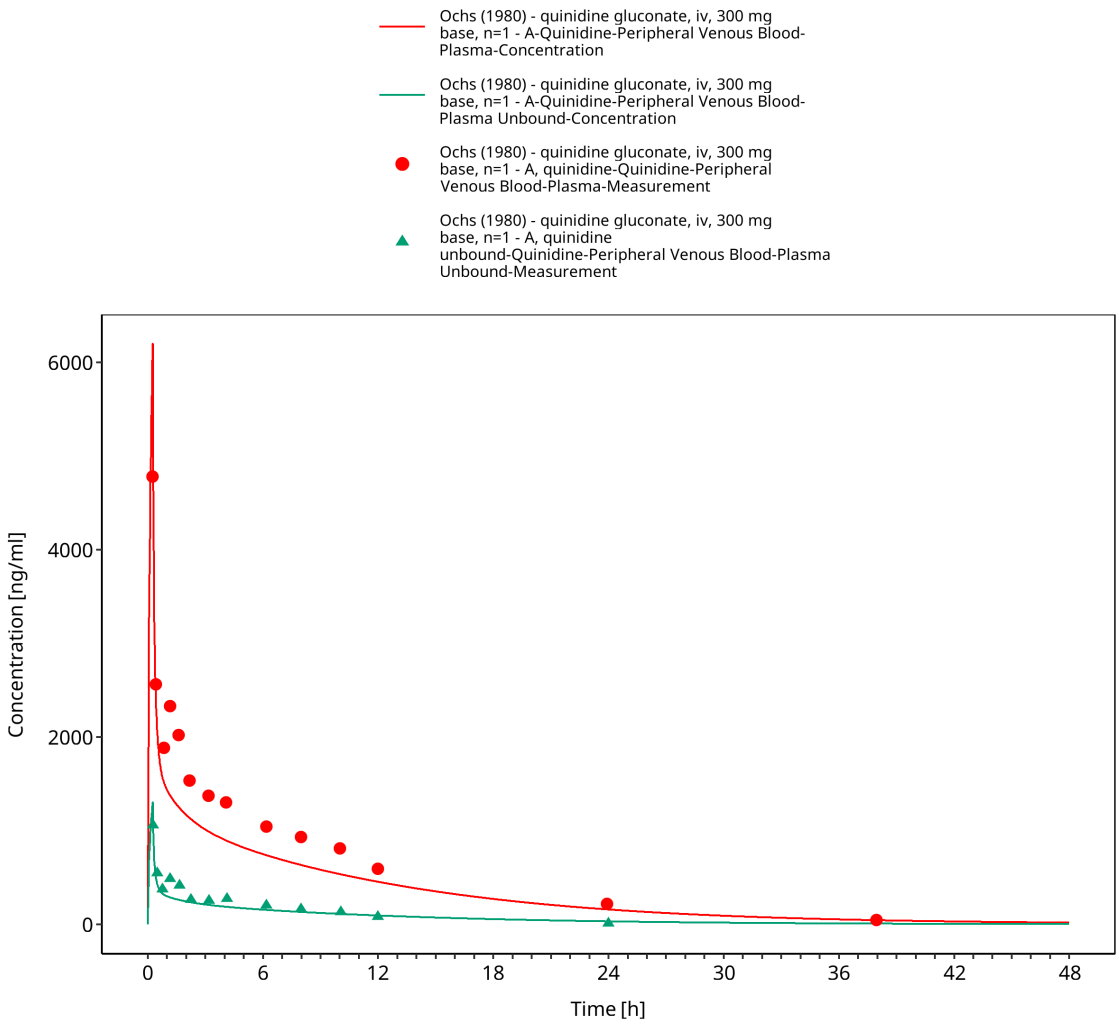


Figure 3-11: Ochs 1980: quinidine gluconate, iv, 300 mg base, n=1 - A

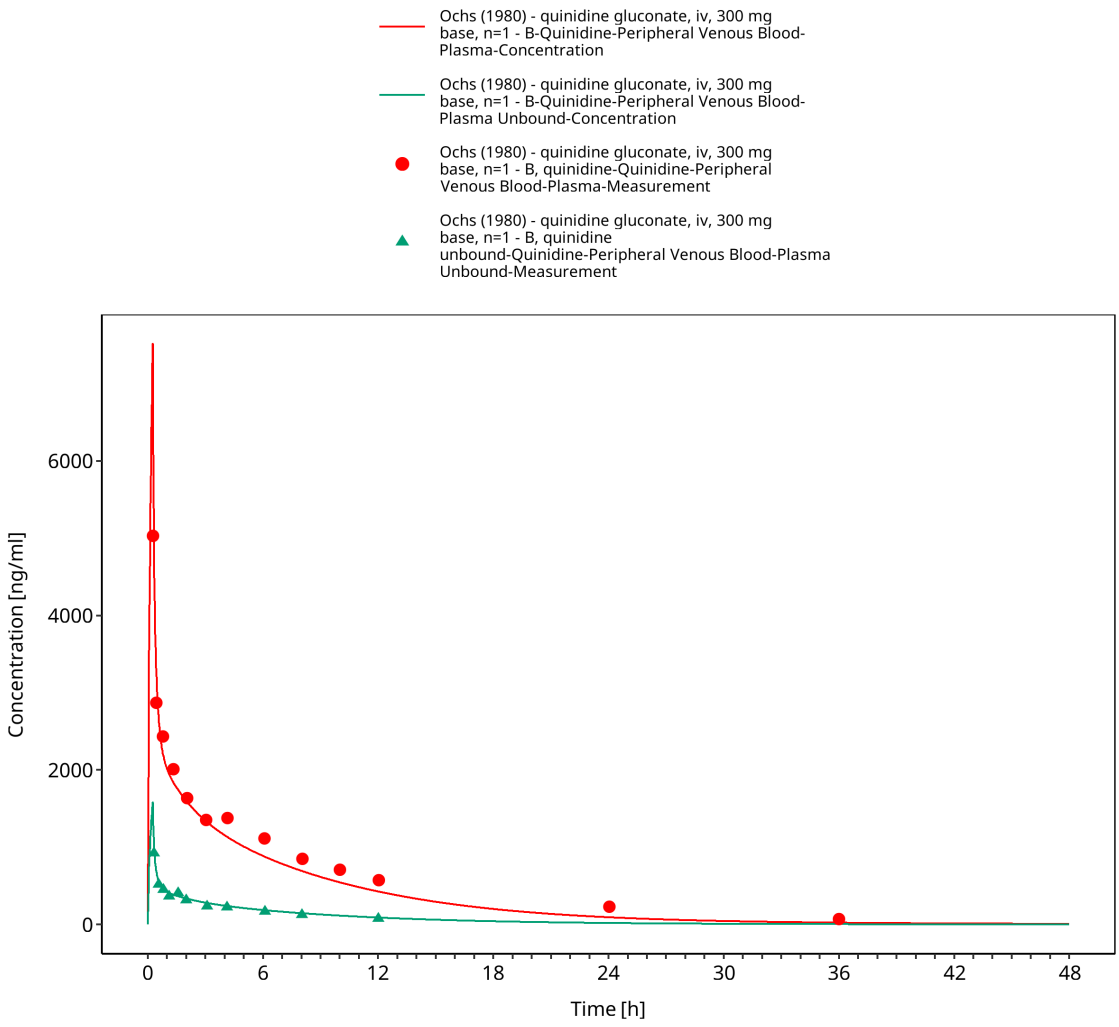


Figure 3-12: Ochs 1980: quinidine gluconate, iv, 300 mg base, n=1 - B

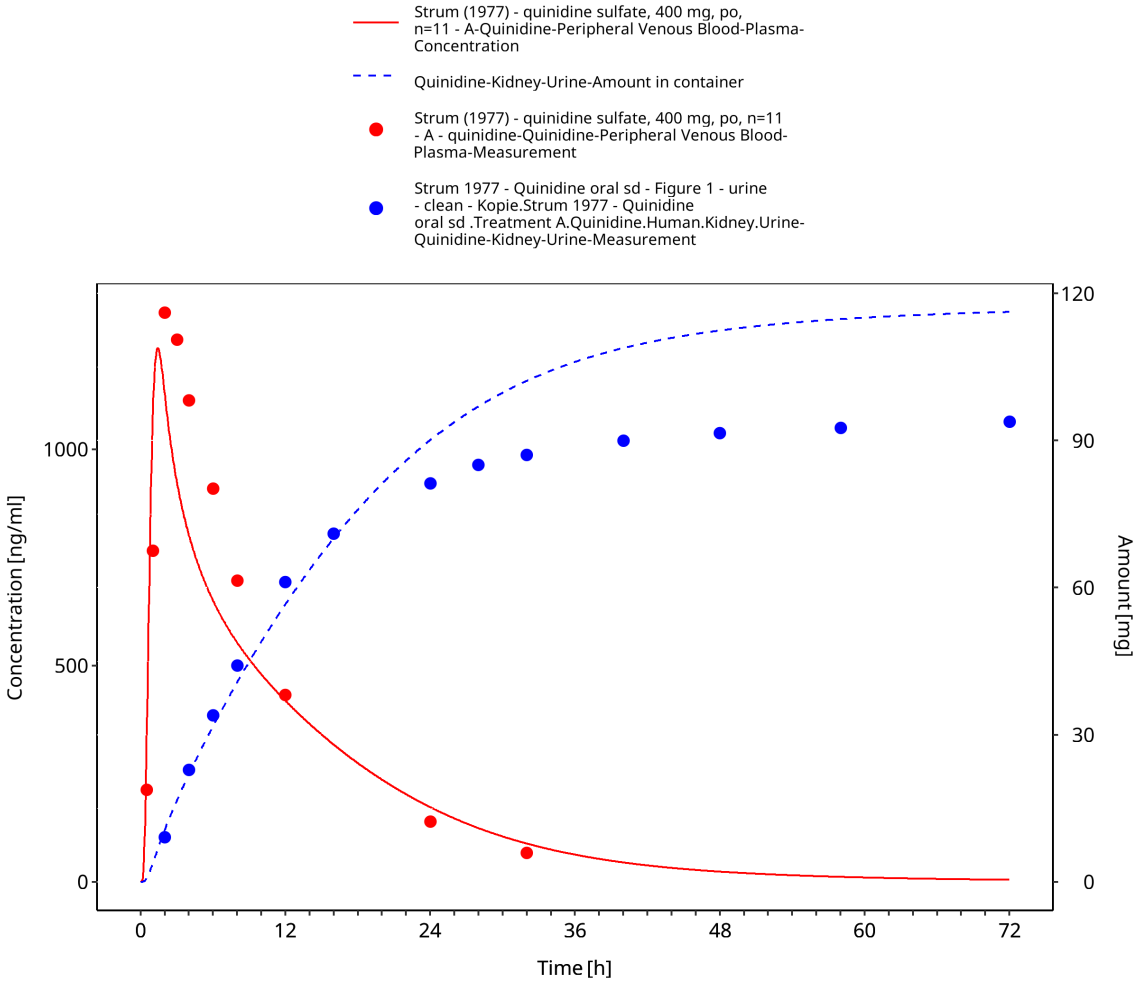


Figure 3-13: Strum 1977: quinidine sulfate, 400 mg, po, n=11 - A

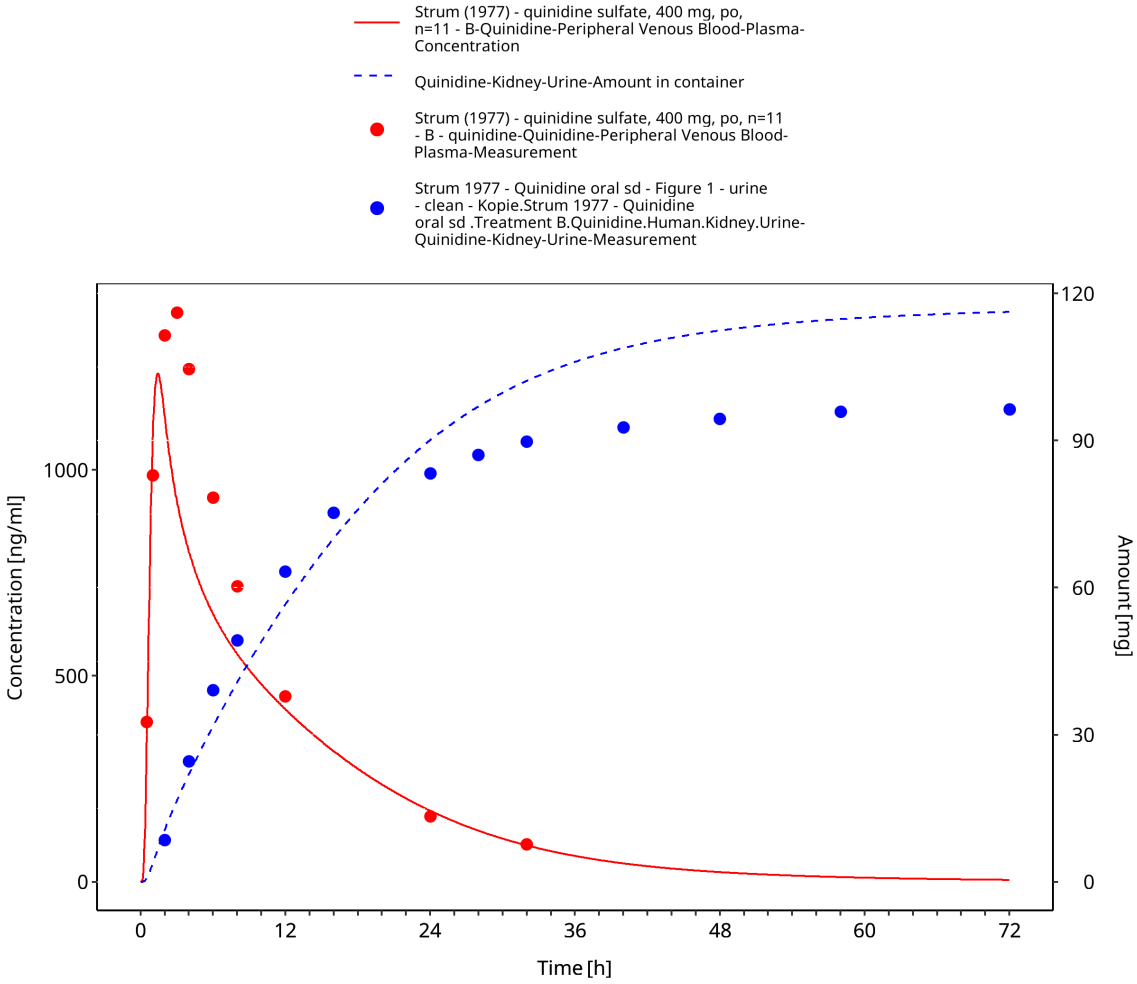


Figure 3-14: Strum 1977: quinidine sulfate, 400 mg, po, n=11 - B

3.3.2 Model Verification

The following verification profiles compare the final model with independent quinidine datasets, including oral and intravenous applications and study-specific interaction conditions. Parent, metabolite, and unbound observations are interpreted as distinct endpoints.

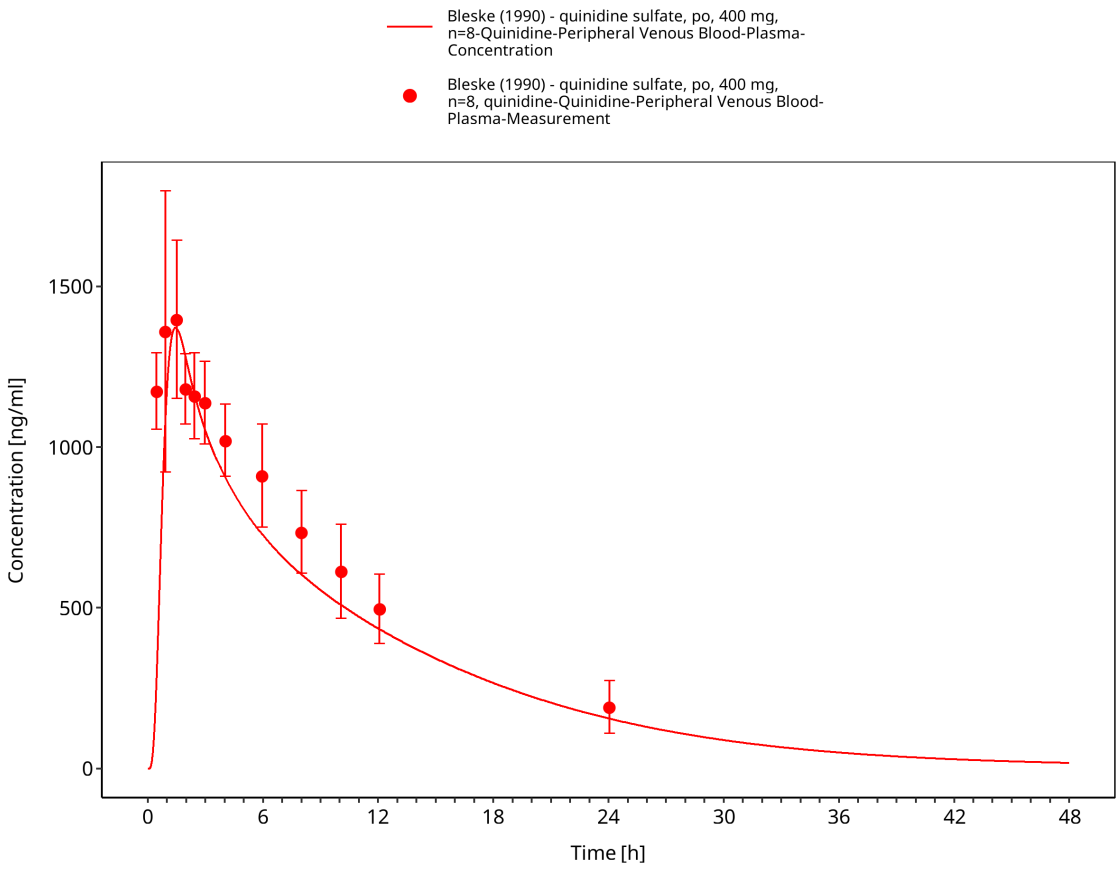


Figure 3-15: Bleske 1990: quinidine sulfate, po, 400 mg, n=8

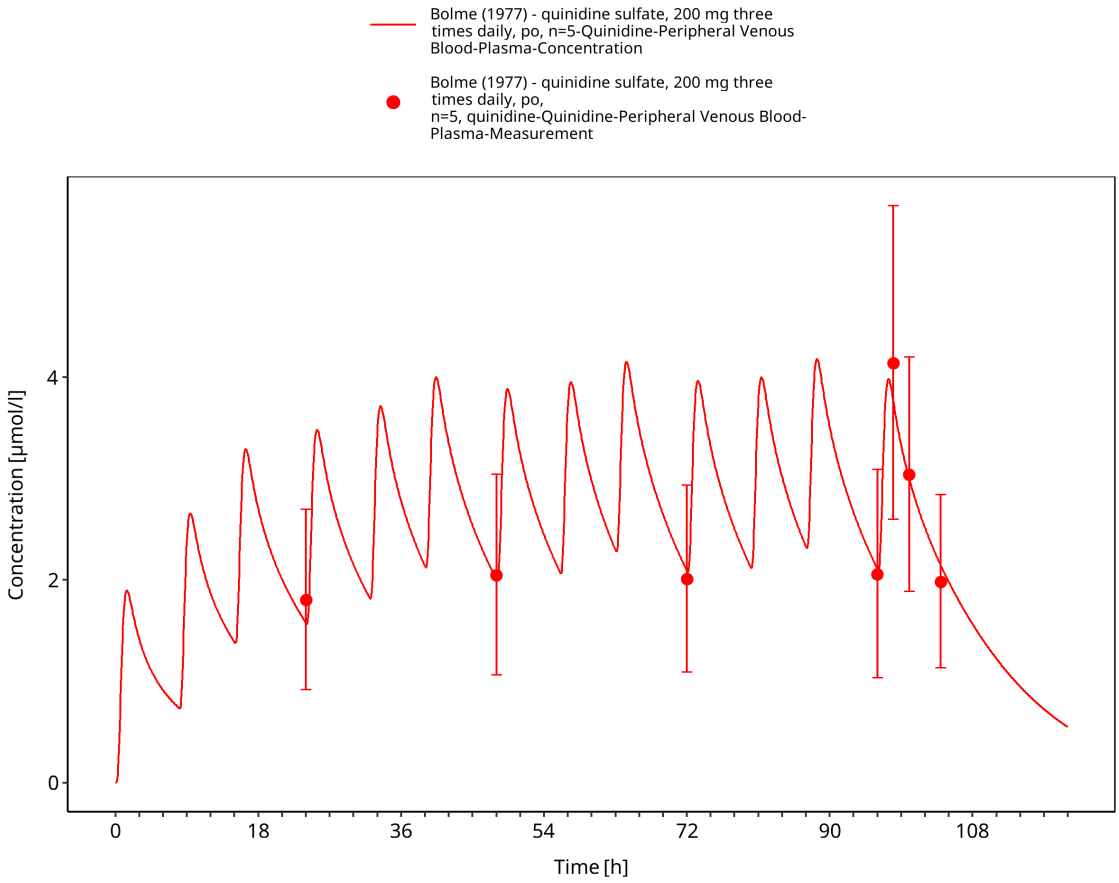


Figure 3-16: Bolme 1977: quinidine sulfate, 200 mg tid, po, n=5

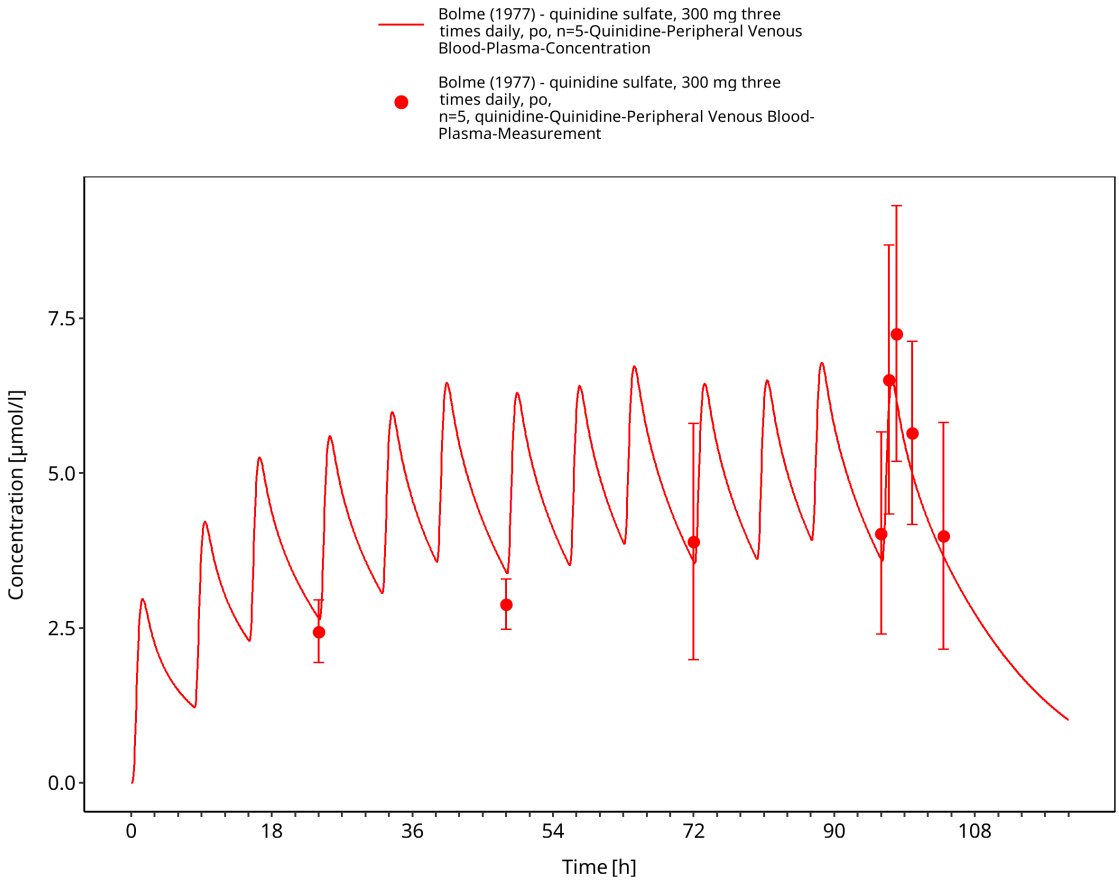


Figure 3-17: Bolme 1977: quinidine sulfate, 300 mg tid, po, n=5

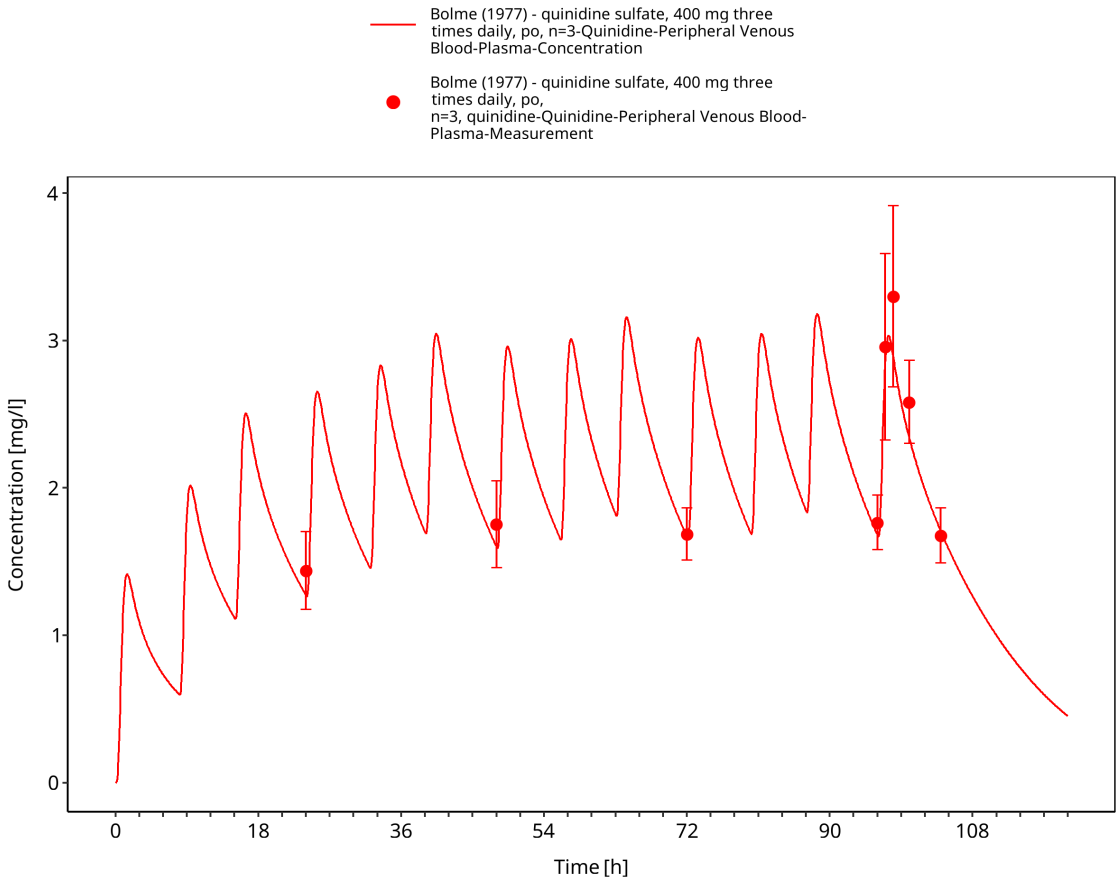


Figure 3-18: Bolme 1977: quinidine sulfate, 400 mg tid, po, n=3

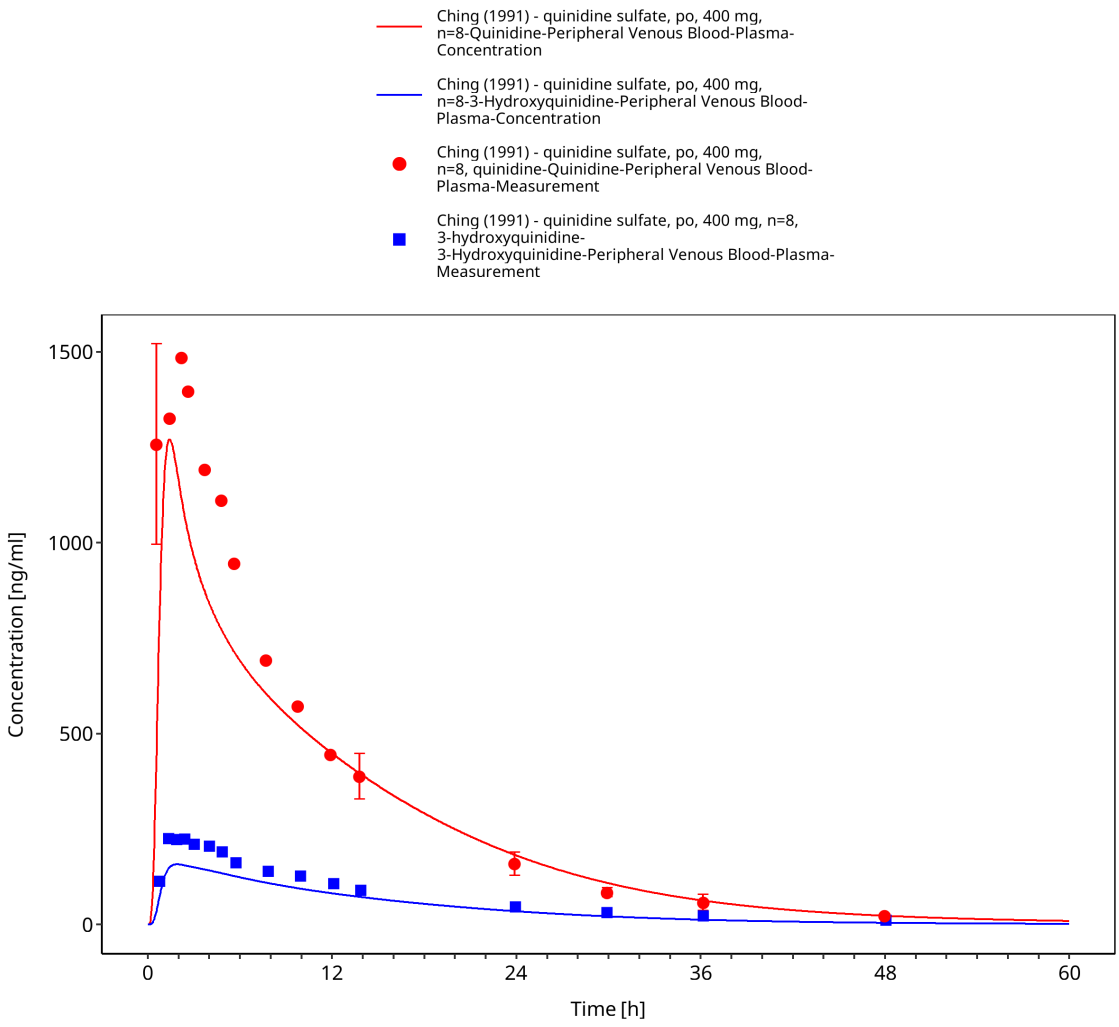


Figure 3-19: Ching 1991: quinidine sulfate, po, 400 mg, n=8

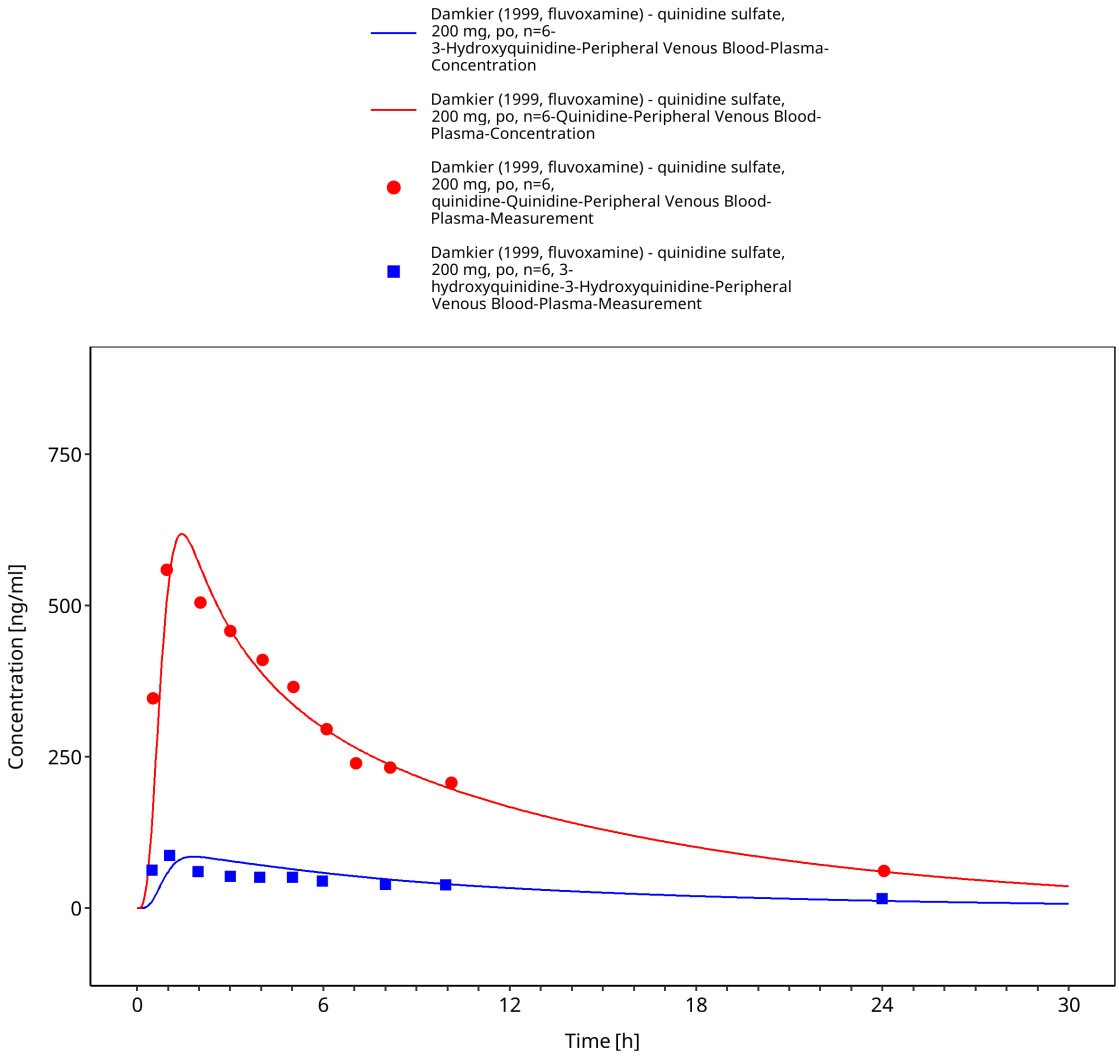


Figure 3-20: Damkier 1999 (fluvoxamine): quinidine sulfate, 200 mg, po, n=6

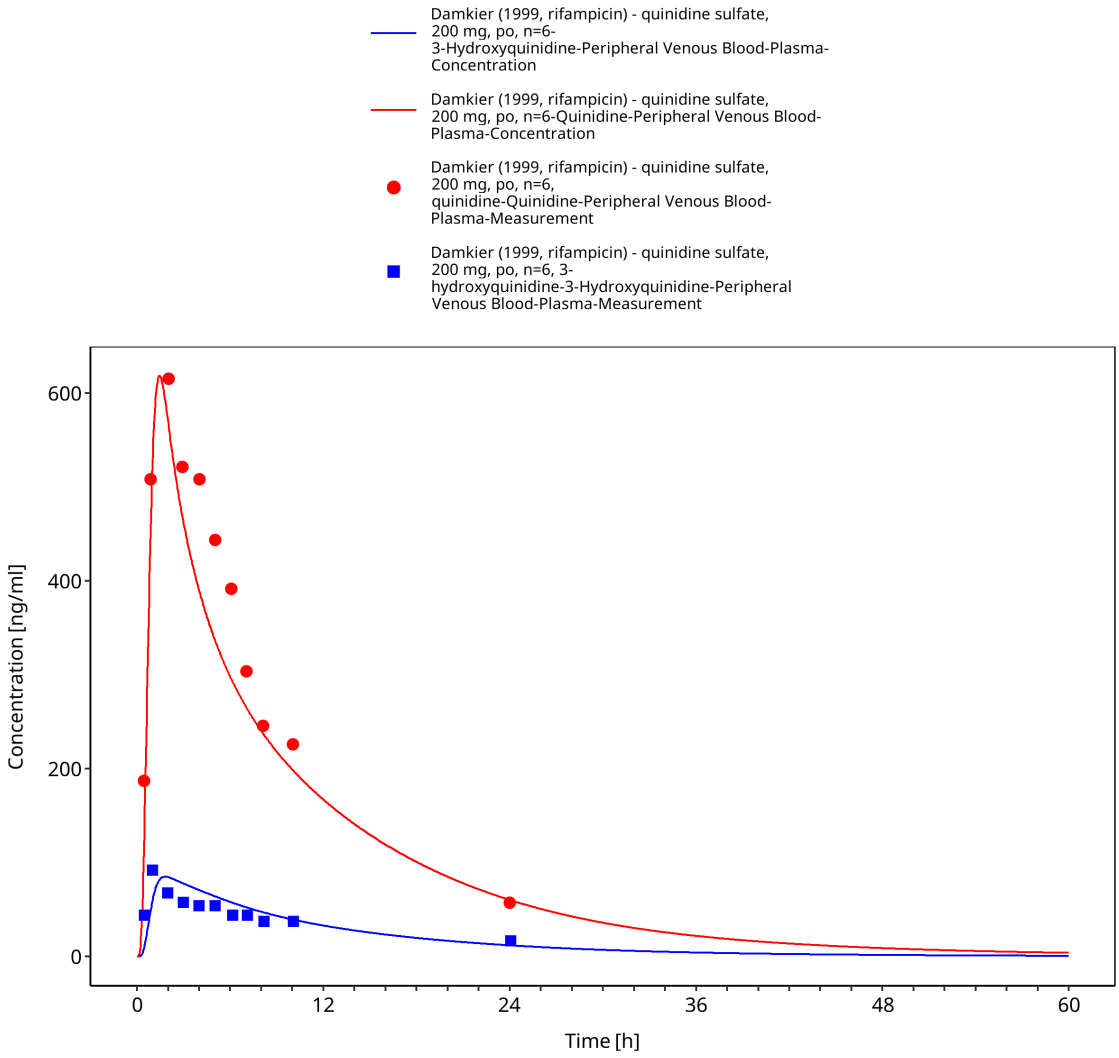


Figure 3-21: Damkier 1999 (rifampicin): quinidine sulfate, 200 mg, po, n=6

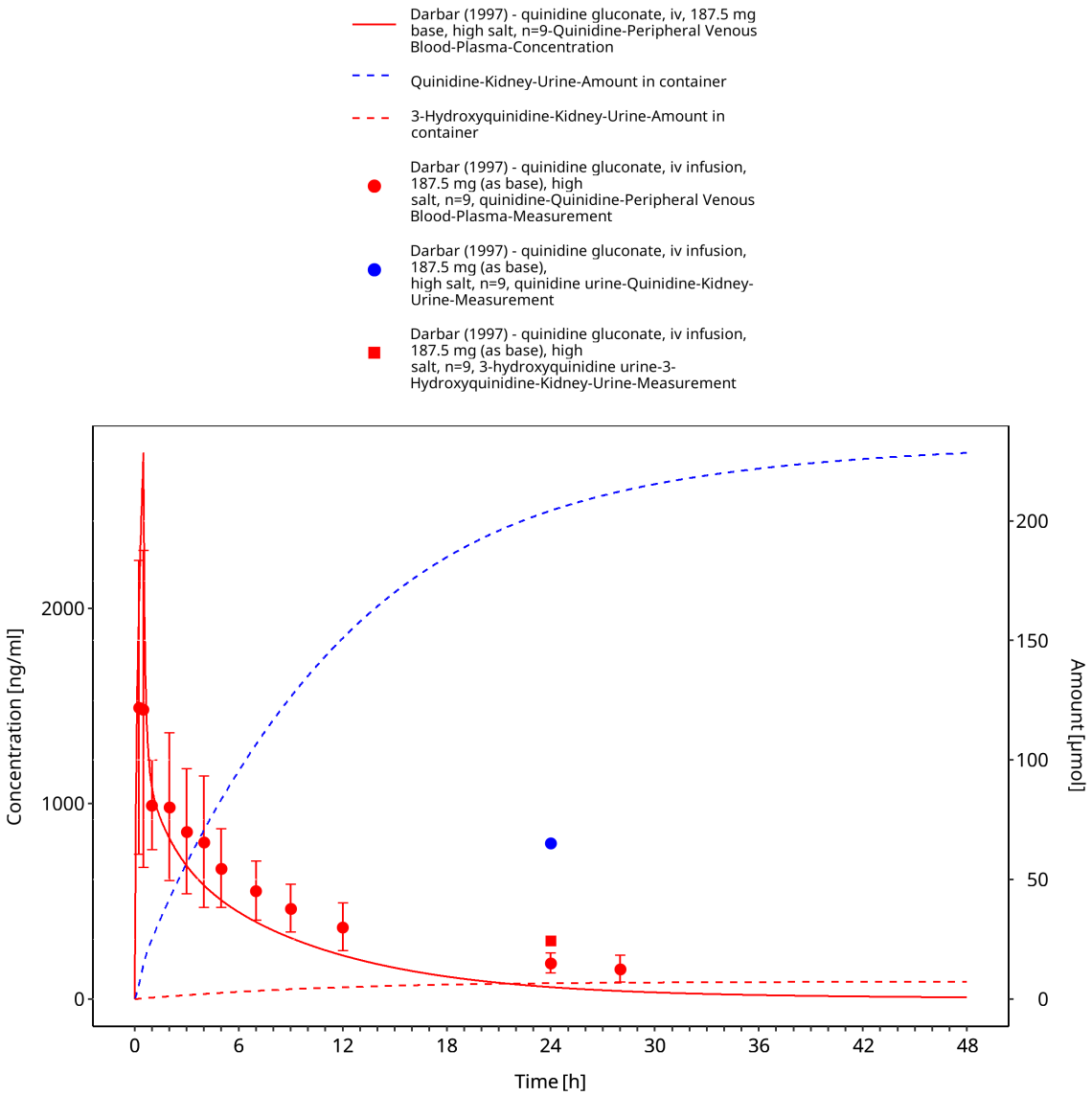


Figure 3-22: Darbar 1997: quinidine gluconate, iv, 187.5 mg base, high salt, n=9

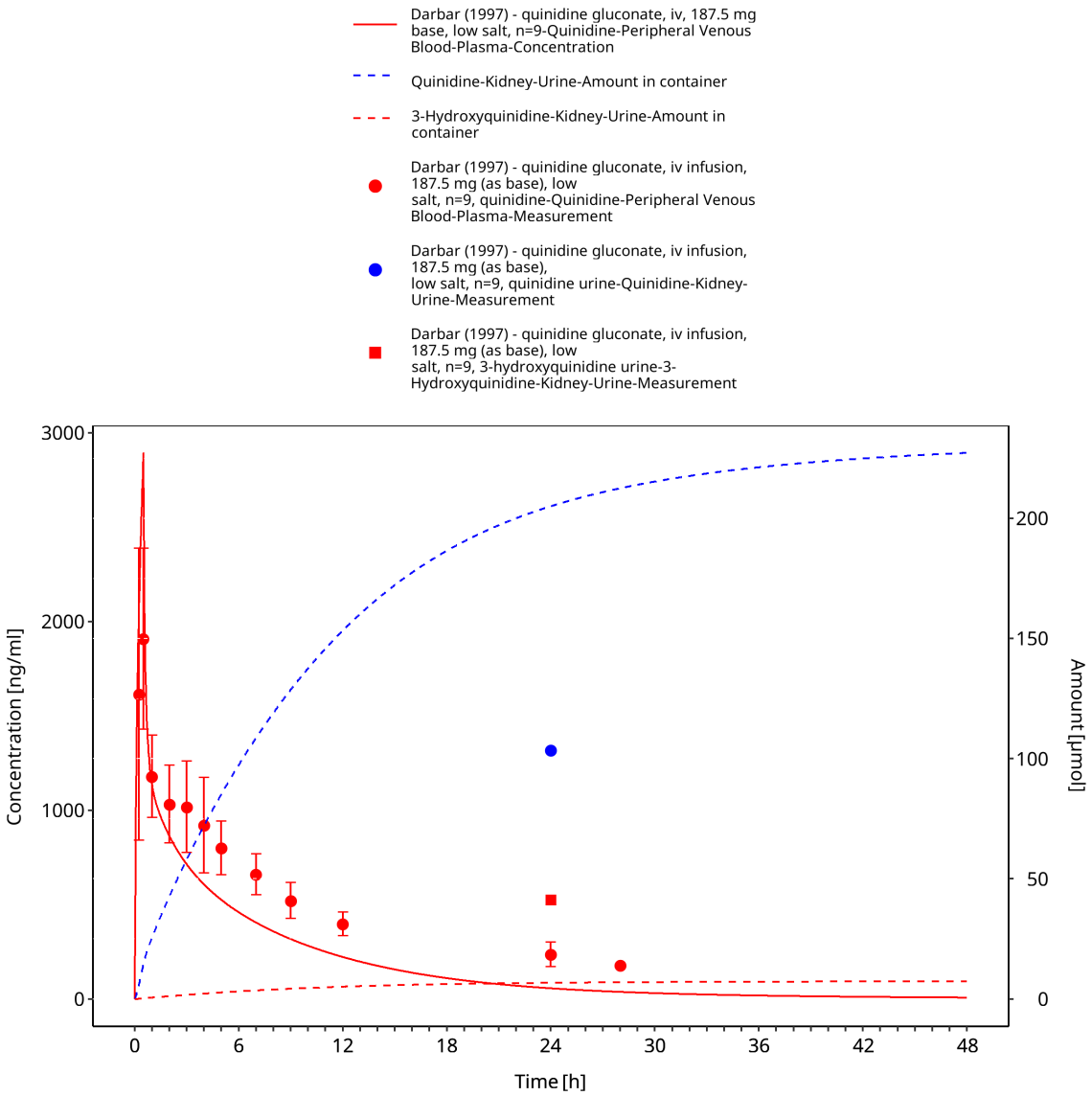


Figure 3-23: Darbar 1997: quinidine gluconate, iv, 187.5 mg base, low salt, n=9

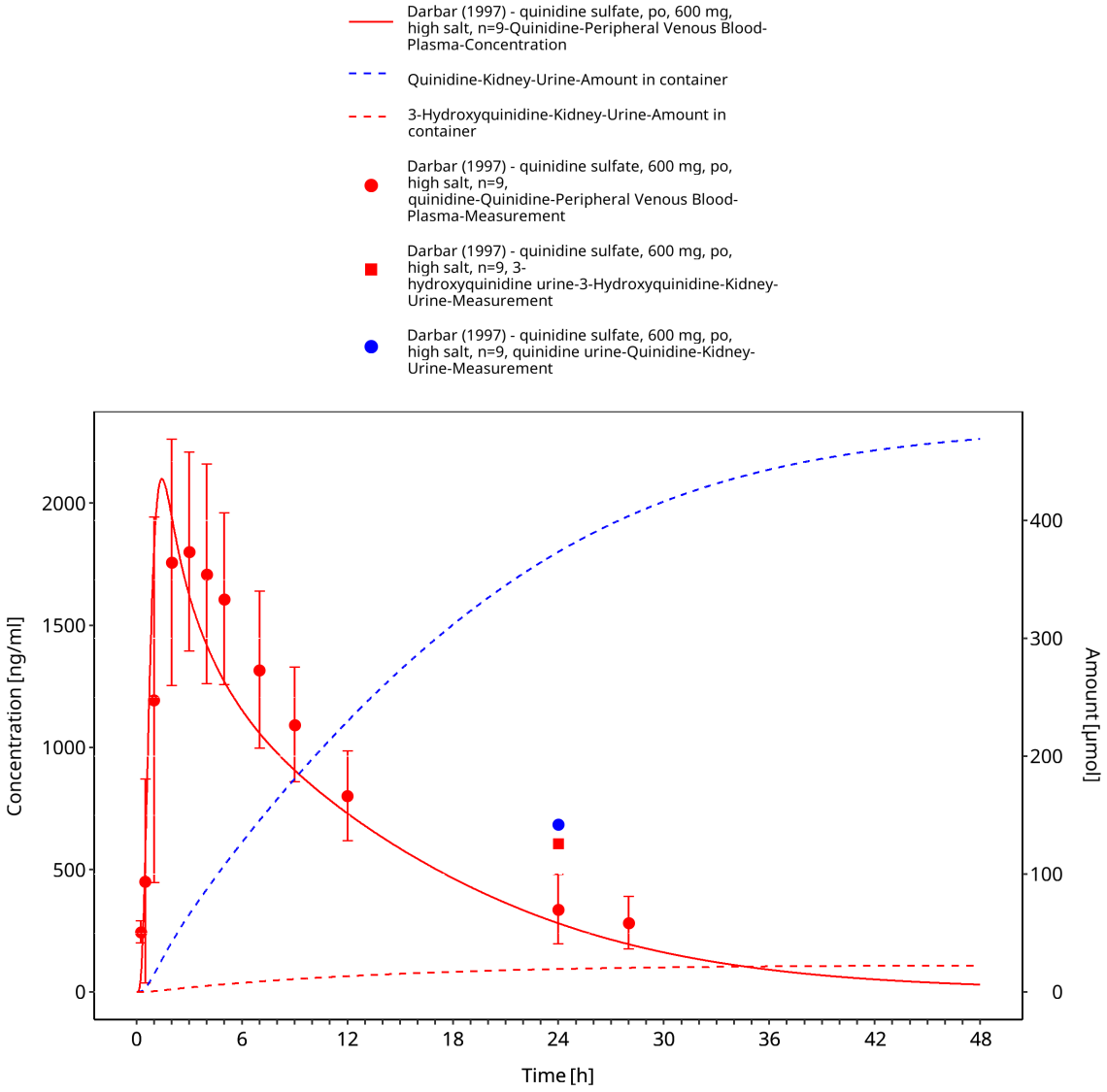


Figure 3-24: Darbar 1997: quinidine sulfate, po, 600 mg, high salt, n=9

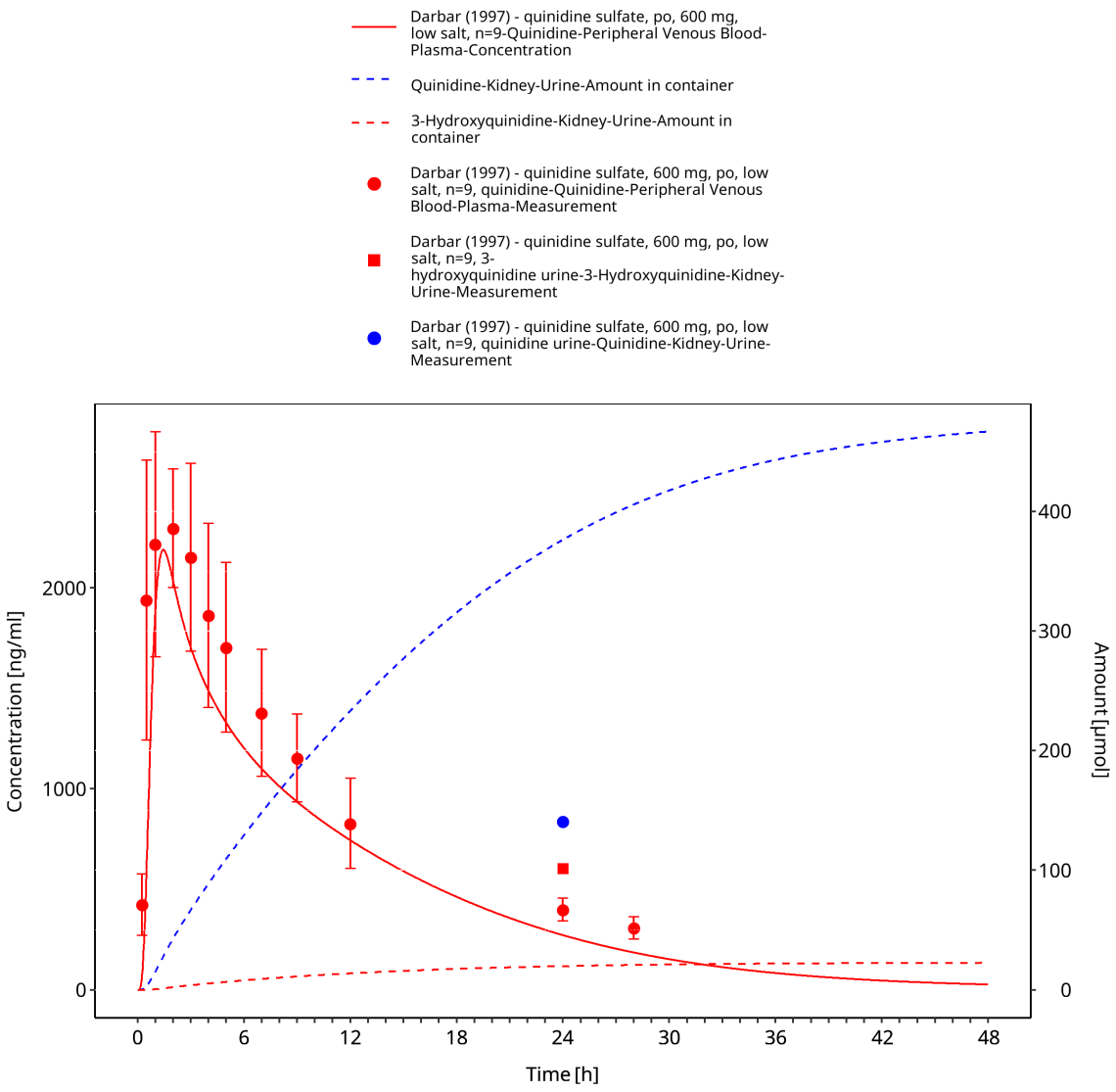


Figure 3-25: Darbar 1997: quinidine sulfate, po, 600 mg, low salt, n=9

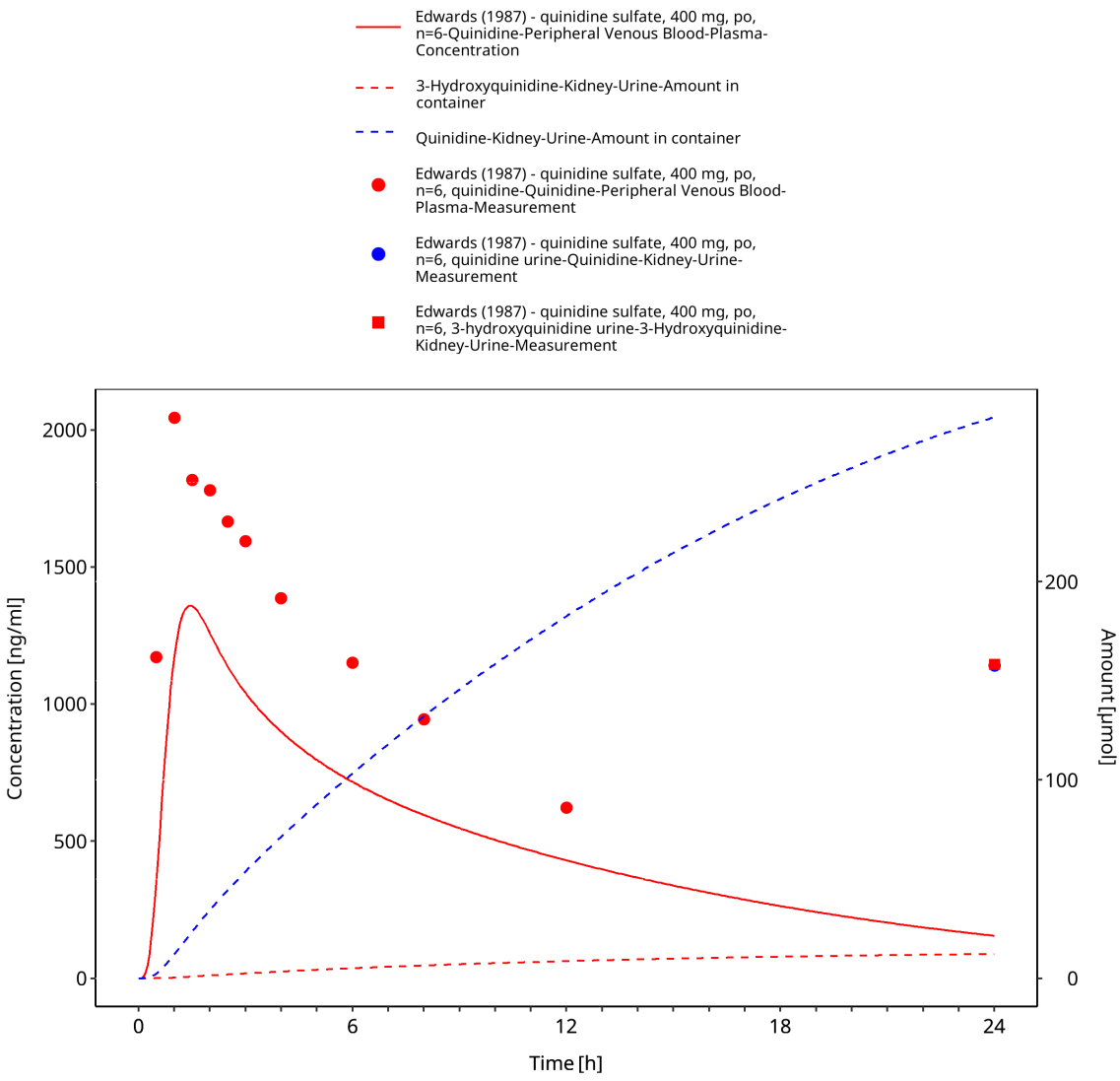


Figure 3-26: Edwards 1987: quinidine sulfate, 400 mg, po, n=6

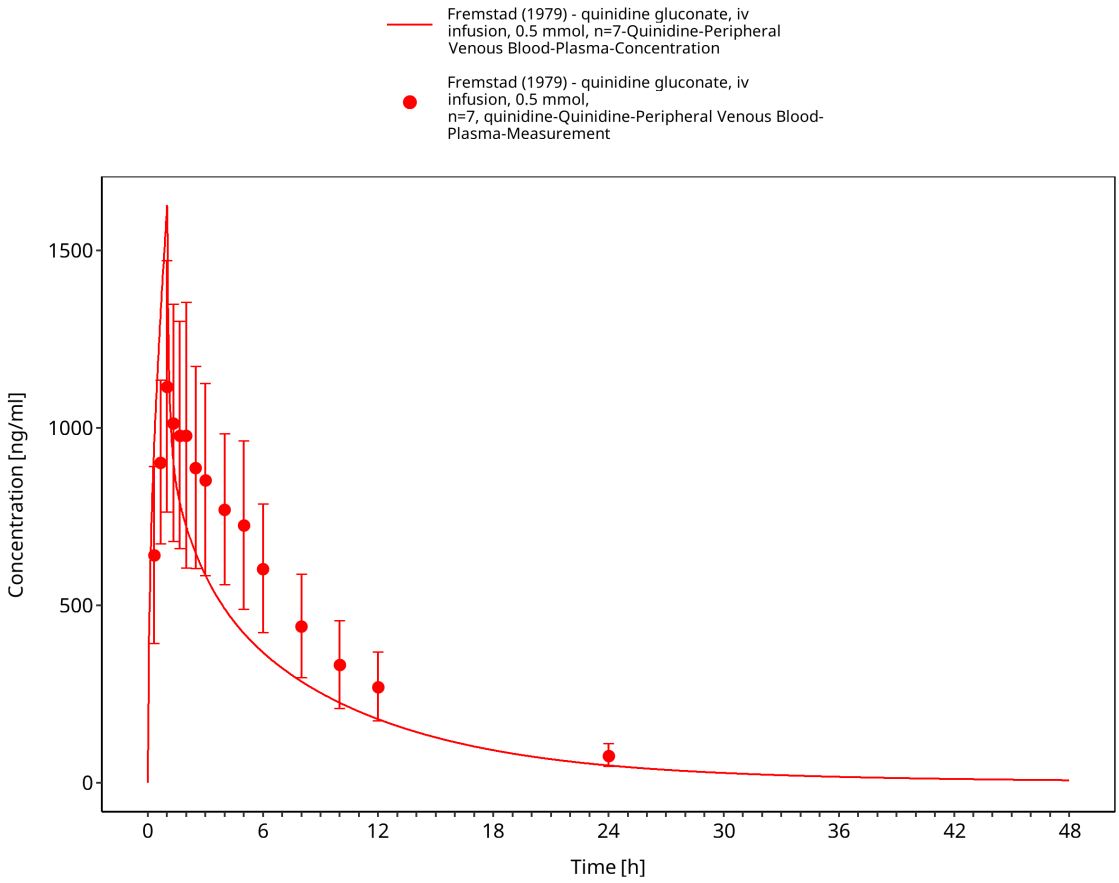


Figure 3-27: Fremstad 1979: quinidine gluconate, iv infusion, 0.5 mmol, n=7

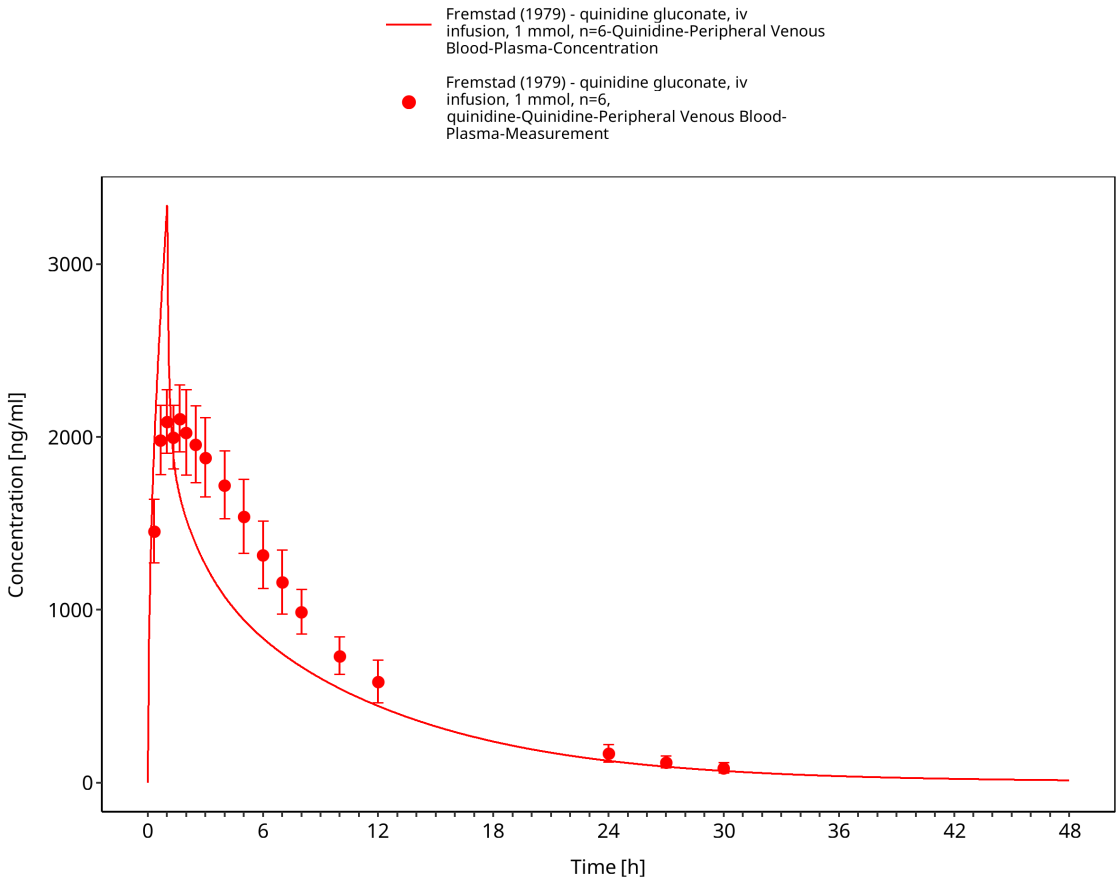


Figure 3-28: Fremstad 1979: quinidine gluconate, iv infusion, 1 mmol, n=6

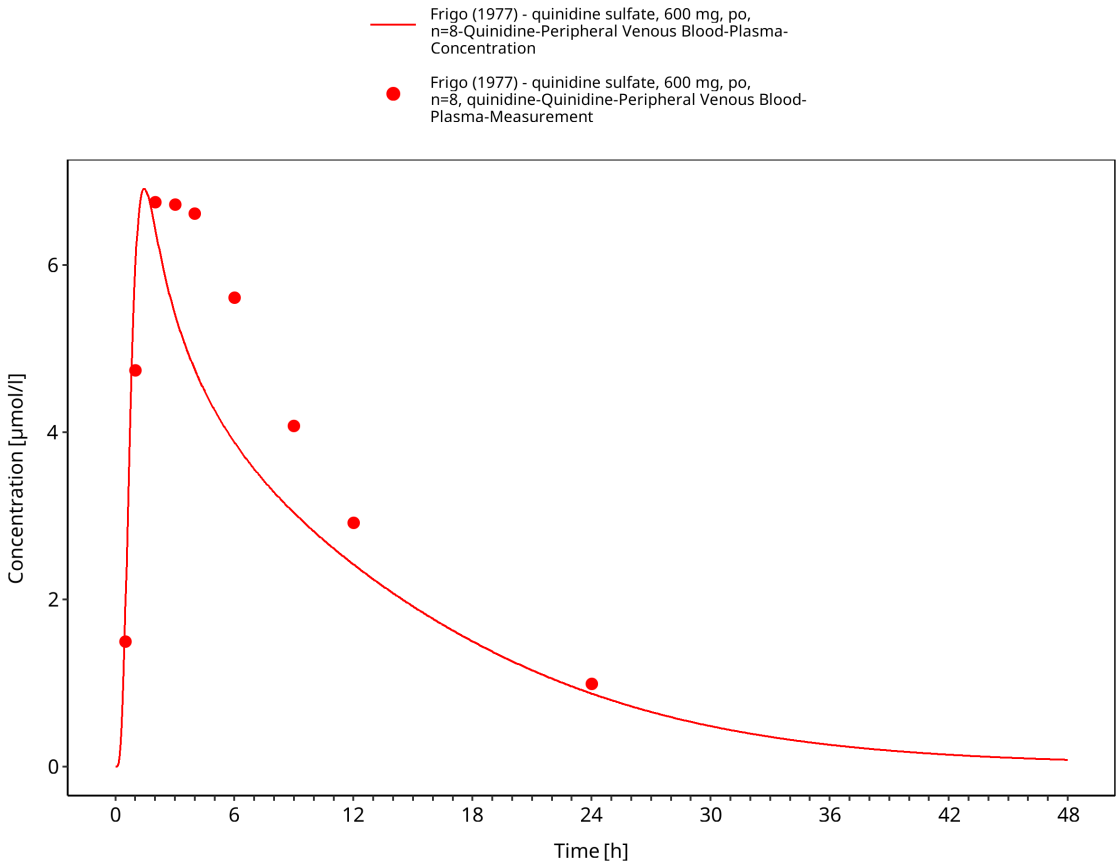


Figure 3-29: Frigo 1977: quinidine sulfate, 600 mg, po, n=8

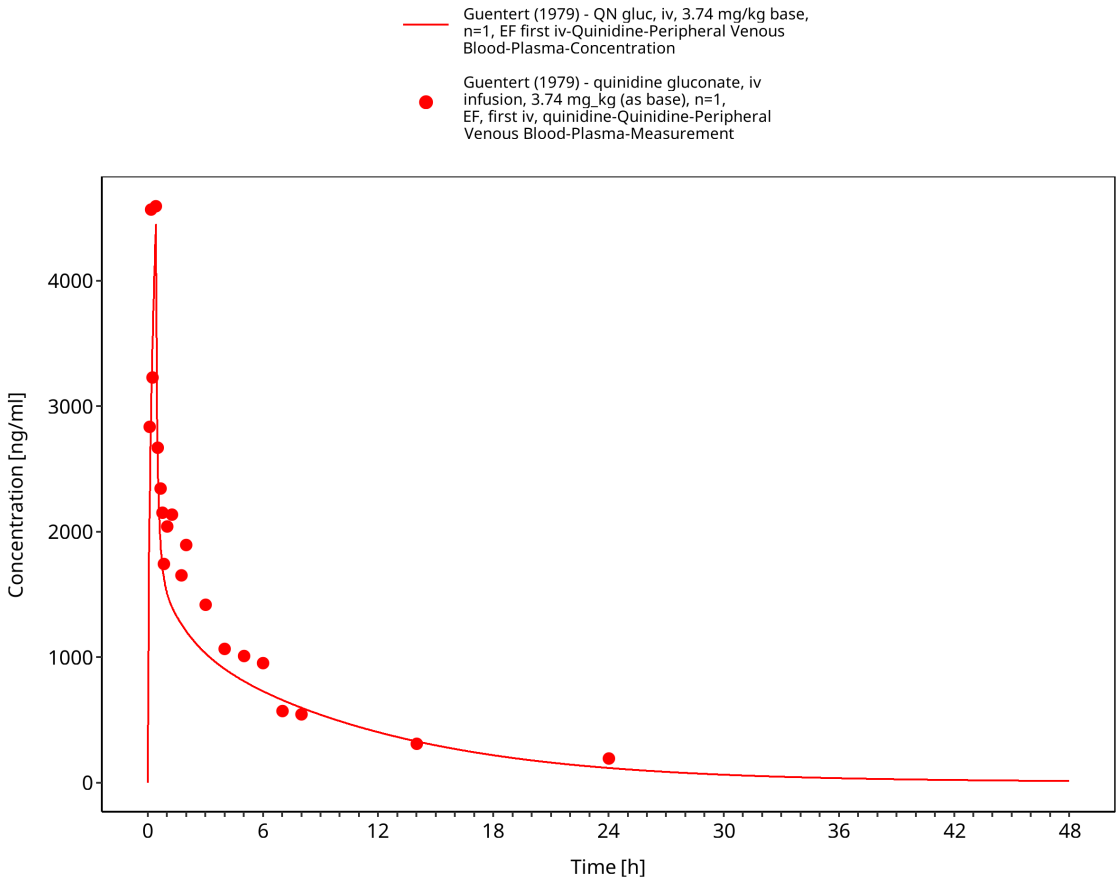


Figure 3-30: Guentert 1979: quinidine gluconate, iv, 3.74 mg/kg base, n=1, EF first iv

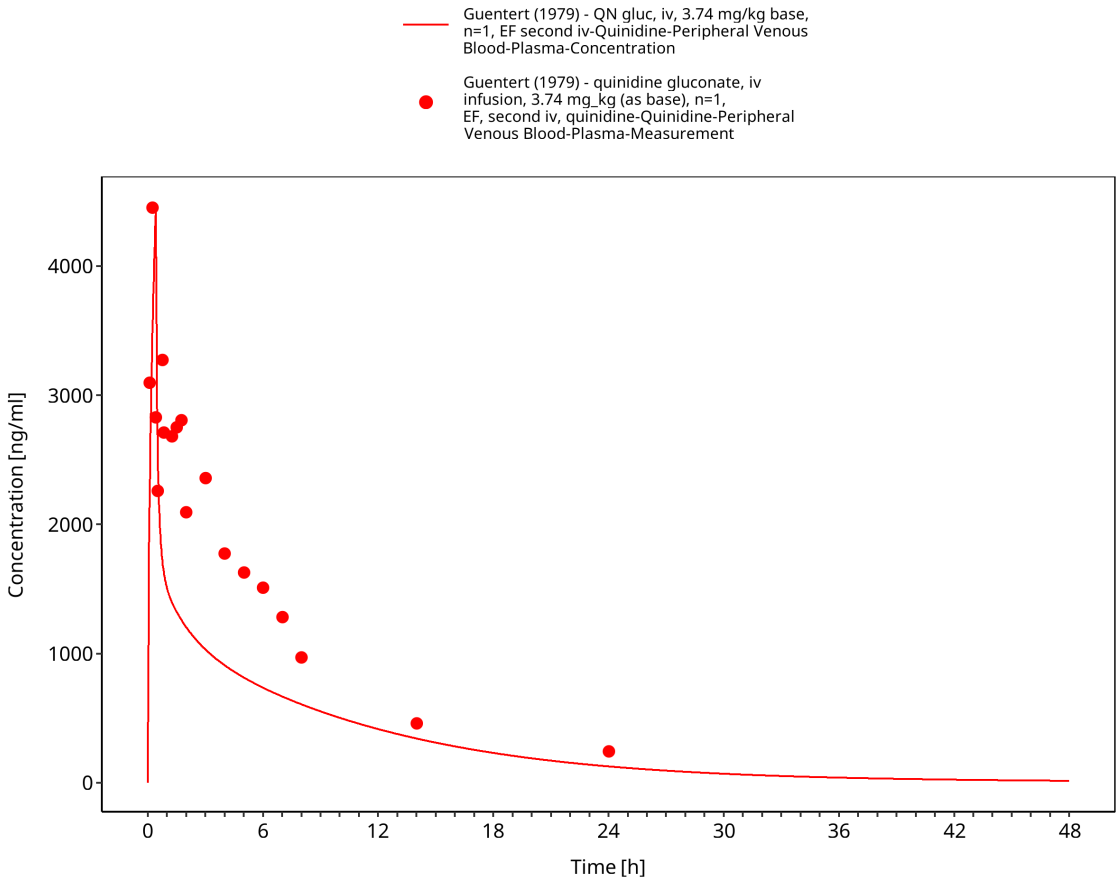


Figure 3-31: Guentert 1979: quinidine gluconate, iv, 3.74 mg/kg base, n=1, EF second iv

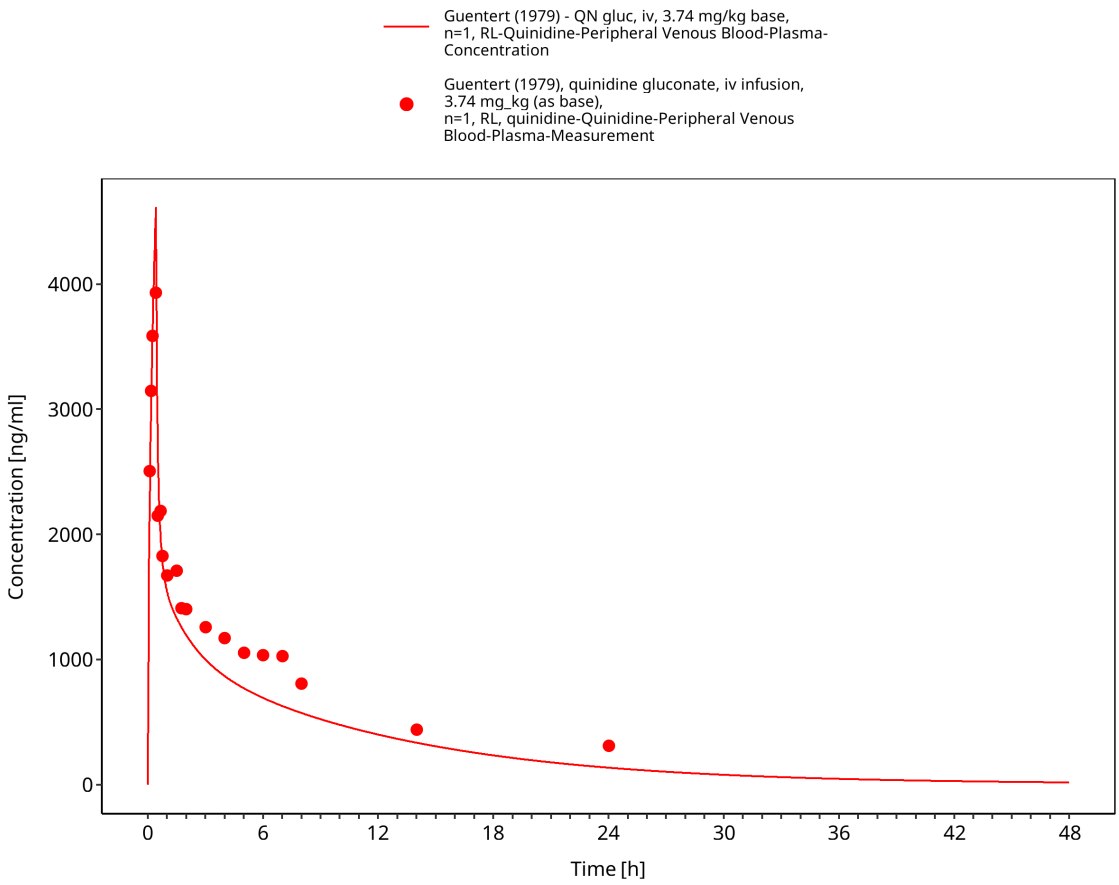


Figure 3-32: Guentert 1979: quinidine gluconate, iv, 3.74 mg/kg base, n=1, RL

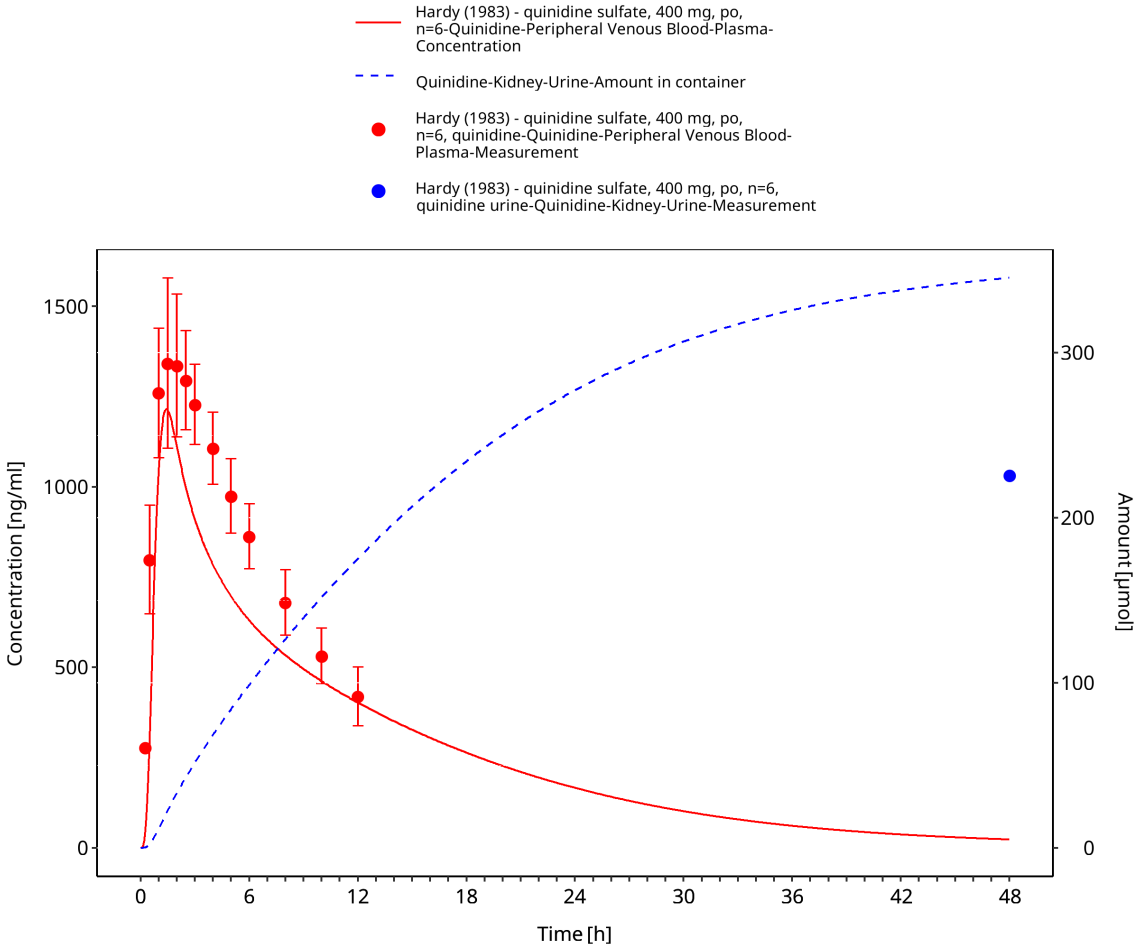


Figure 3-33: Hardy 1983: quinidine sulfate, 400 mg, po, n=6

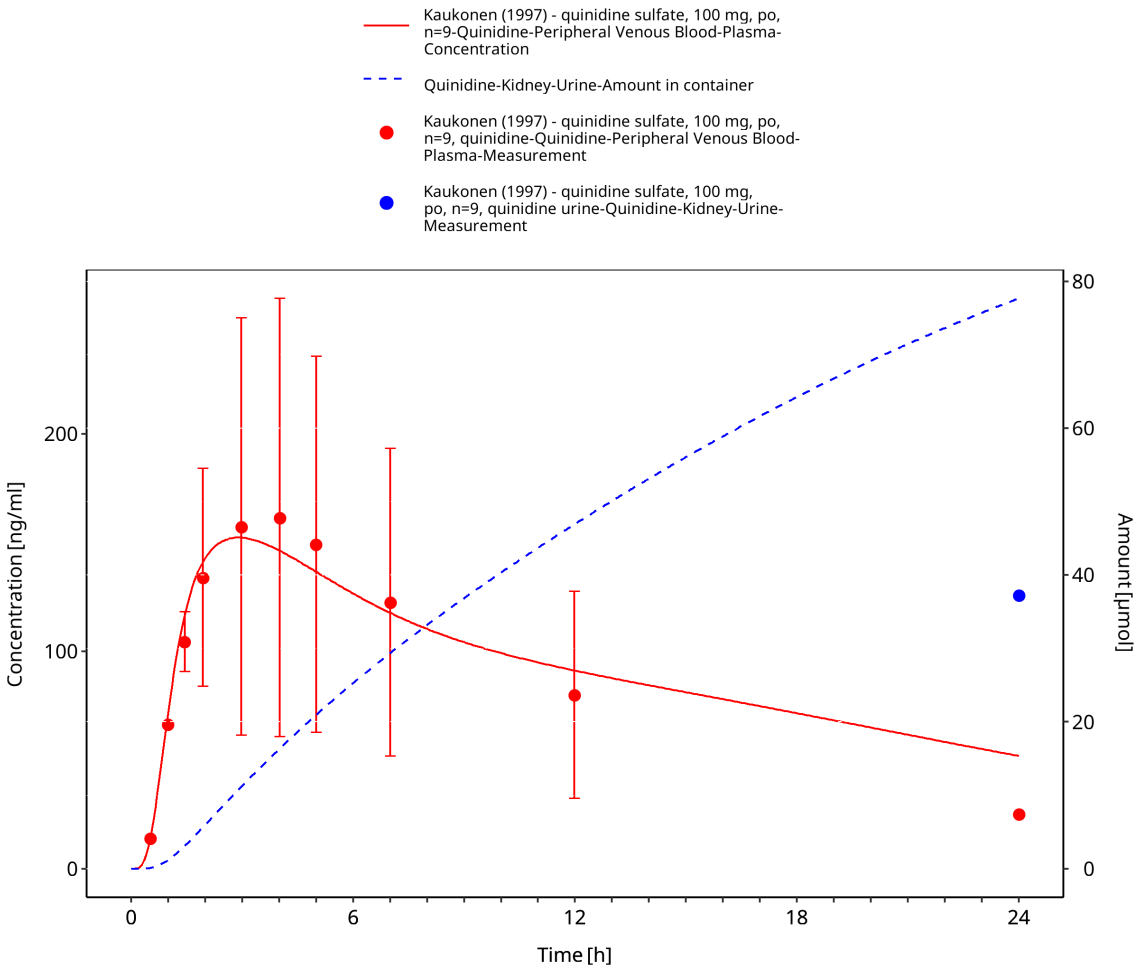


Figure 3-34: Kaukonen 1997: quinidine sulfate, 100 mg, po, n=9

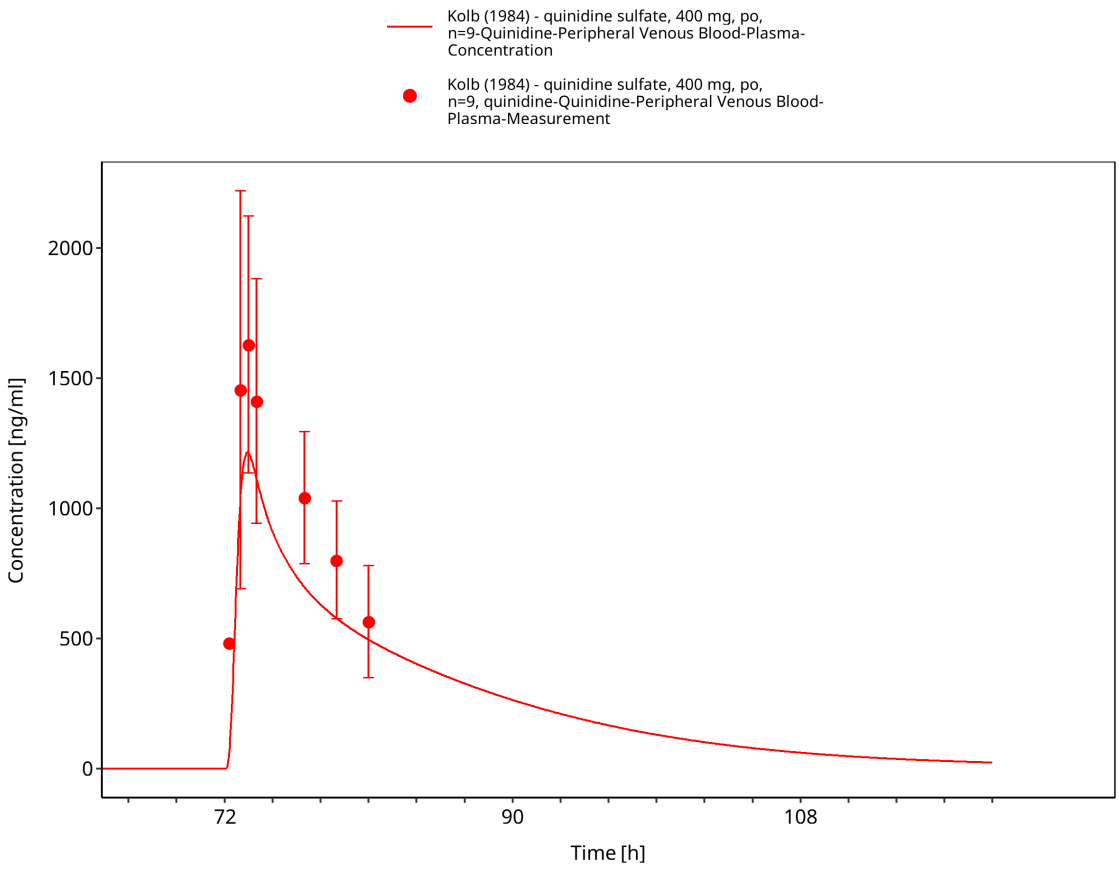


Figure 3-35: Kolb 1984: quinidine sulfate, 400 mg, po, n=9

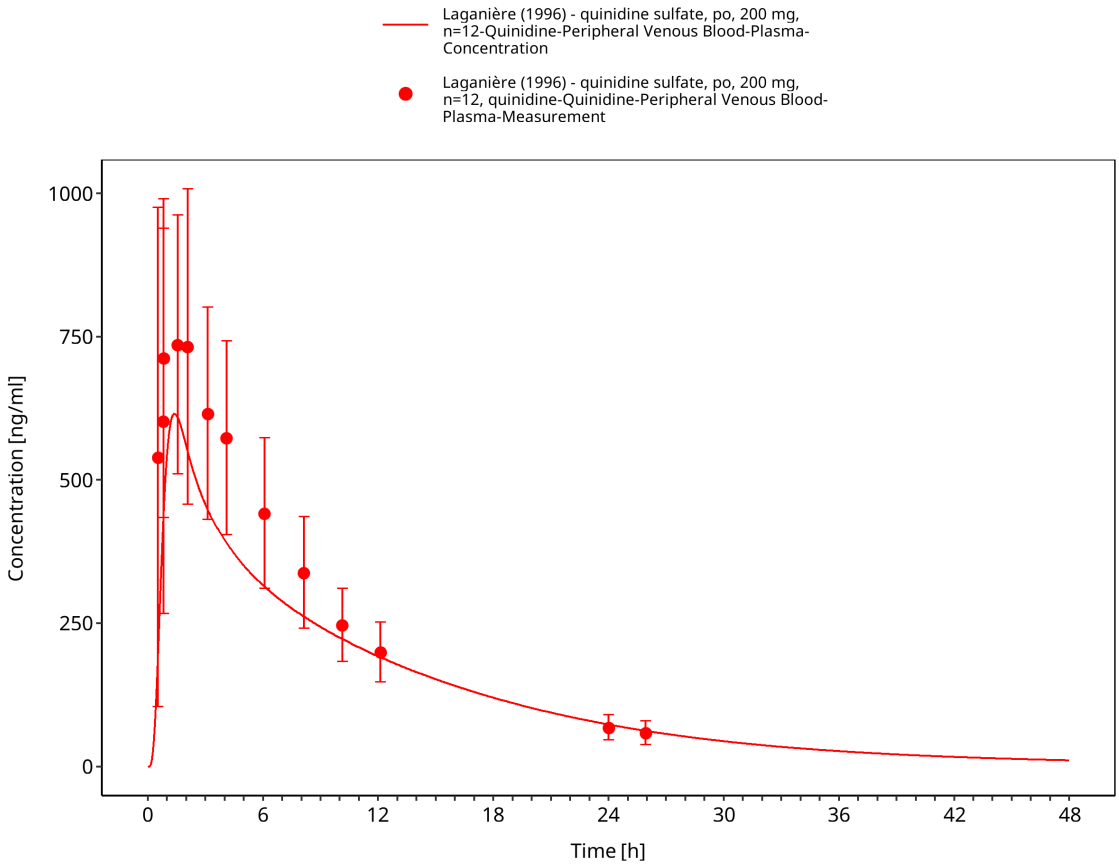


Figure 3-36: Laganière 1996: quinidine sulfate, po, 200 mg, n=12

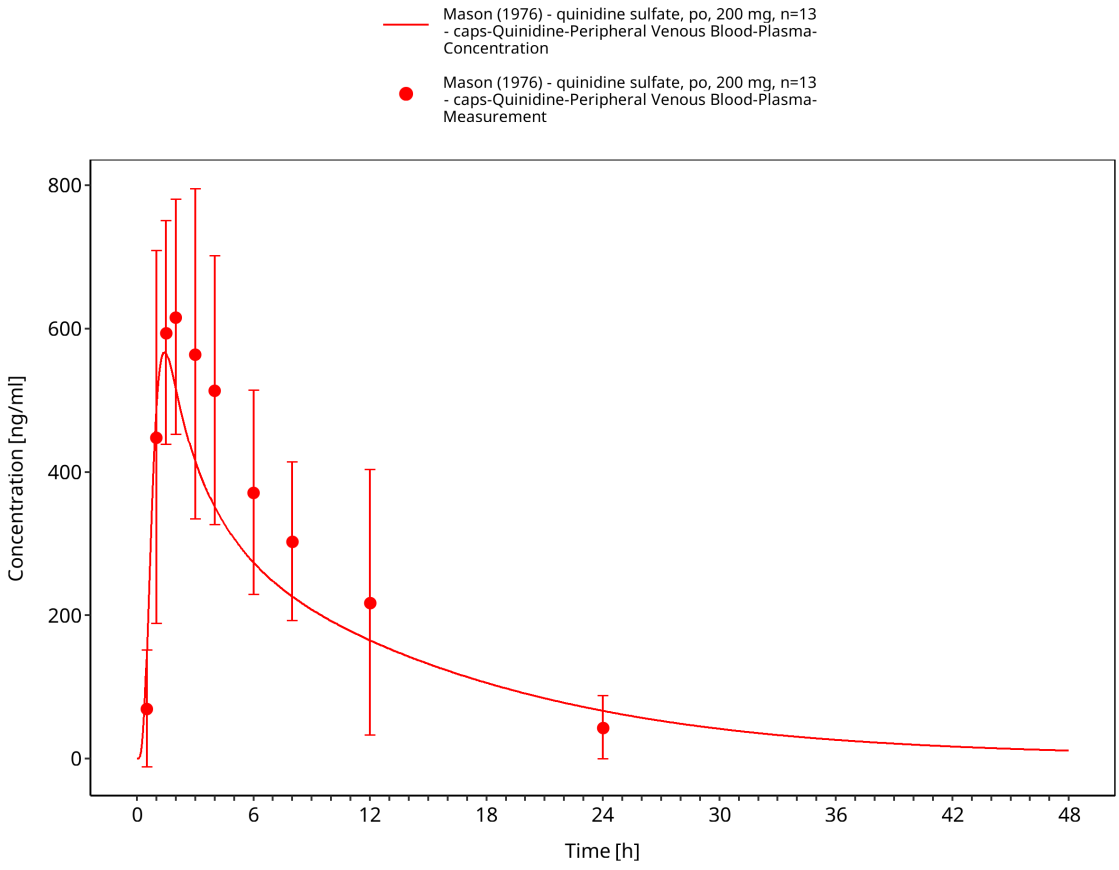


Figure 3-37: Mason 1976: quinidine sulfate, po, 200 mg, n=13 - caps

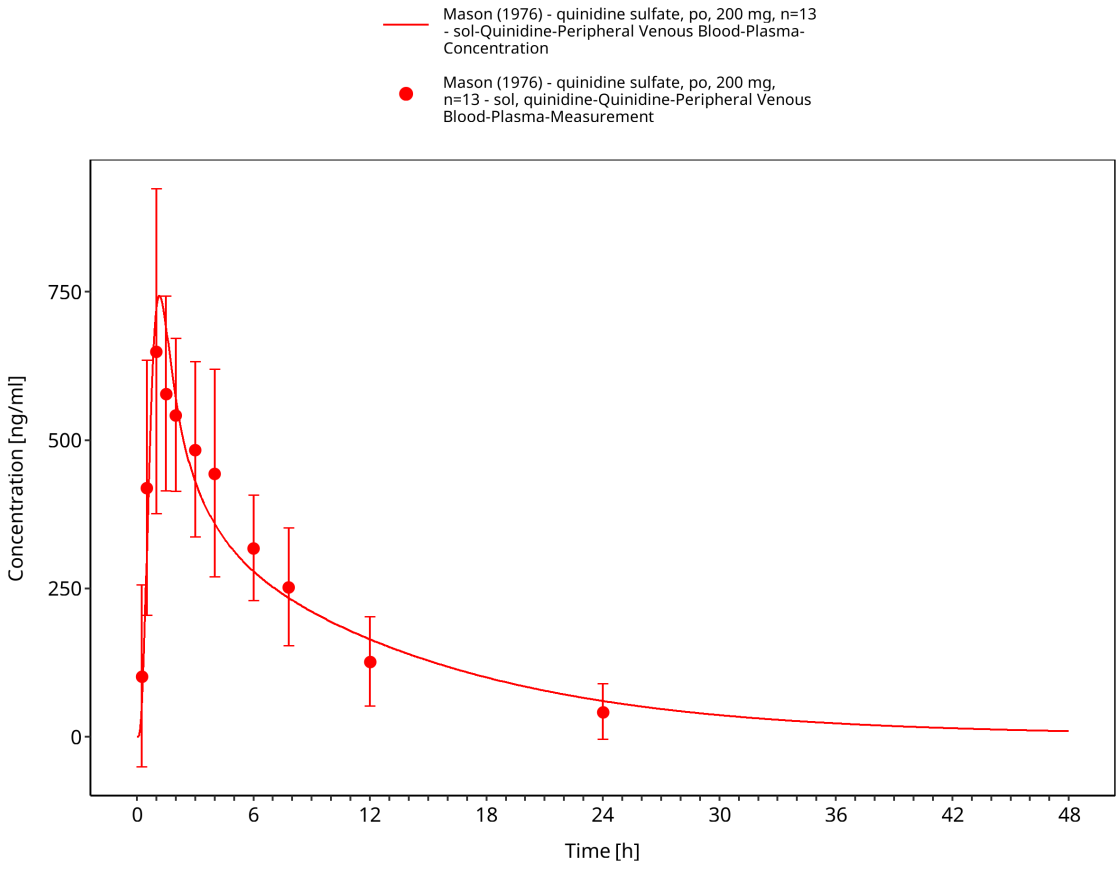


Figure 3-38: Mason 1976: quinidine sulfate, po, 200 mg, n=13 - sol

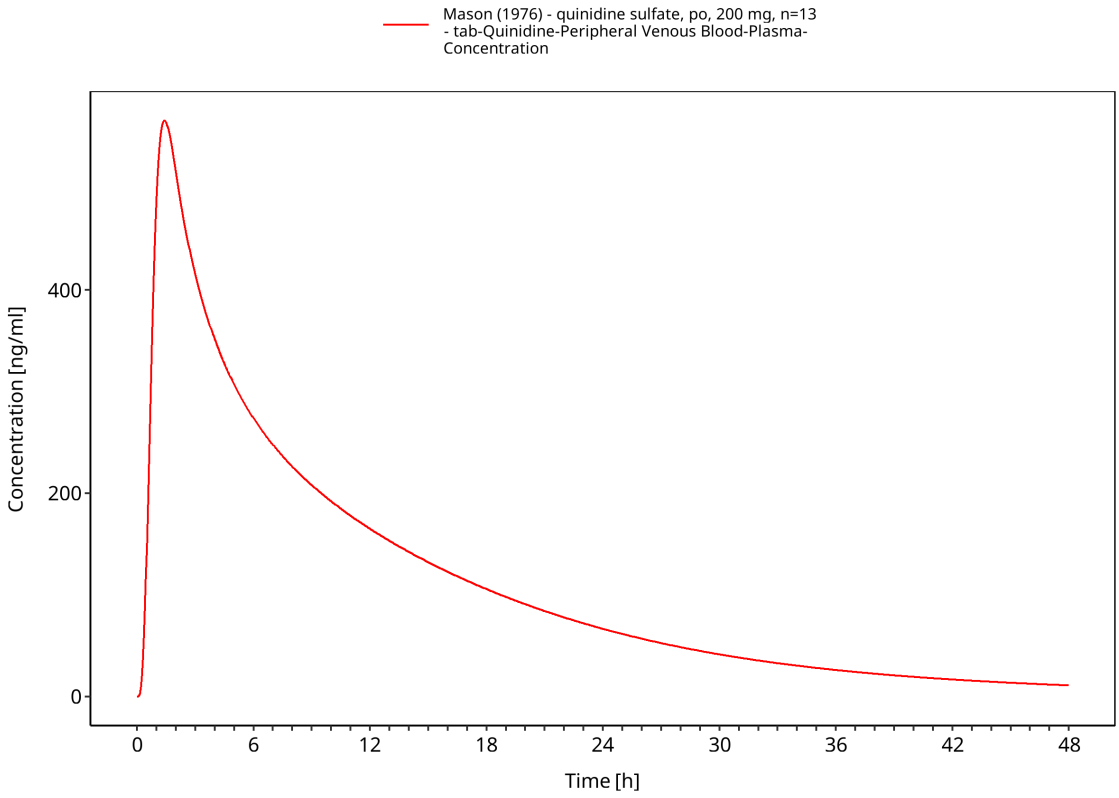


Figure 3-39: Mason 1976: quinidine sulfate, po, 200 mg, n=13 - tab

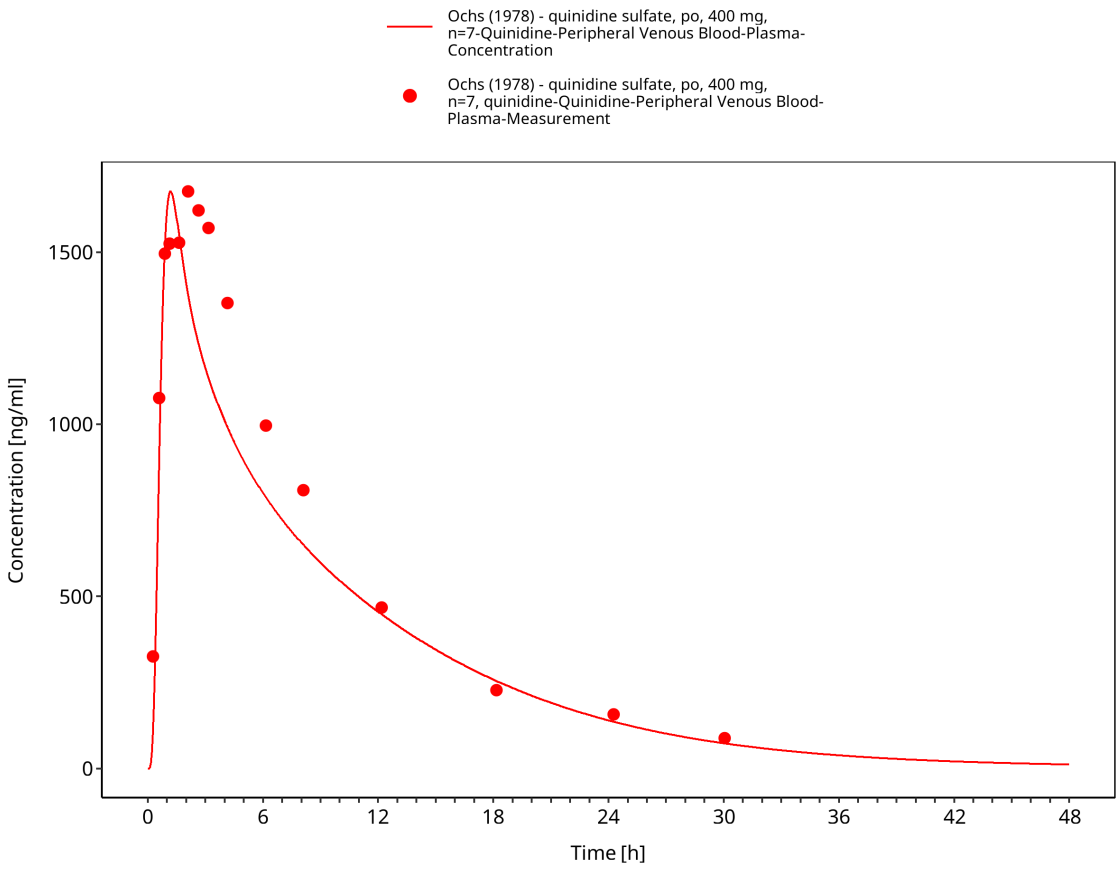


Figure 3-40: Ochs 1978: quinidine sulfate, po, 400 mg, n=7

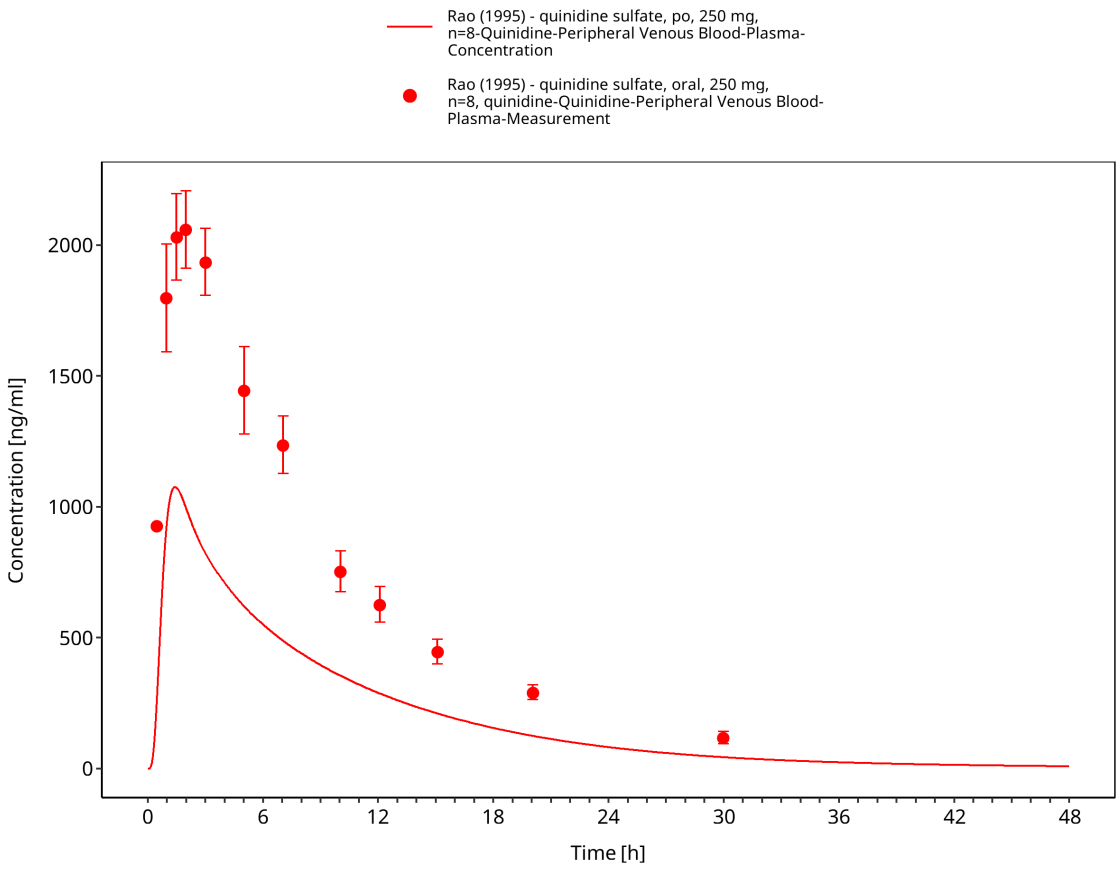


Figure 3-41: Rao 1995: quinidine sulfate, po, 250 mg, n=8

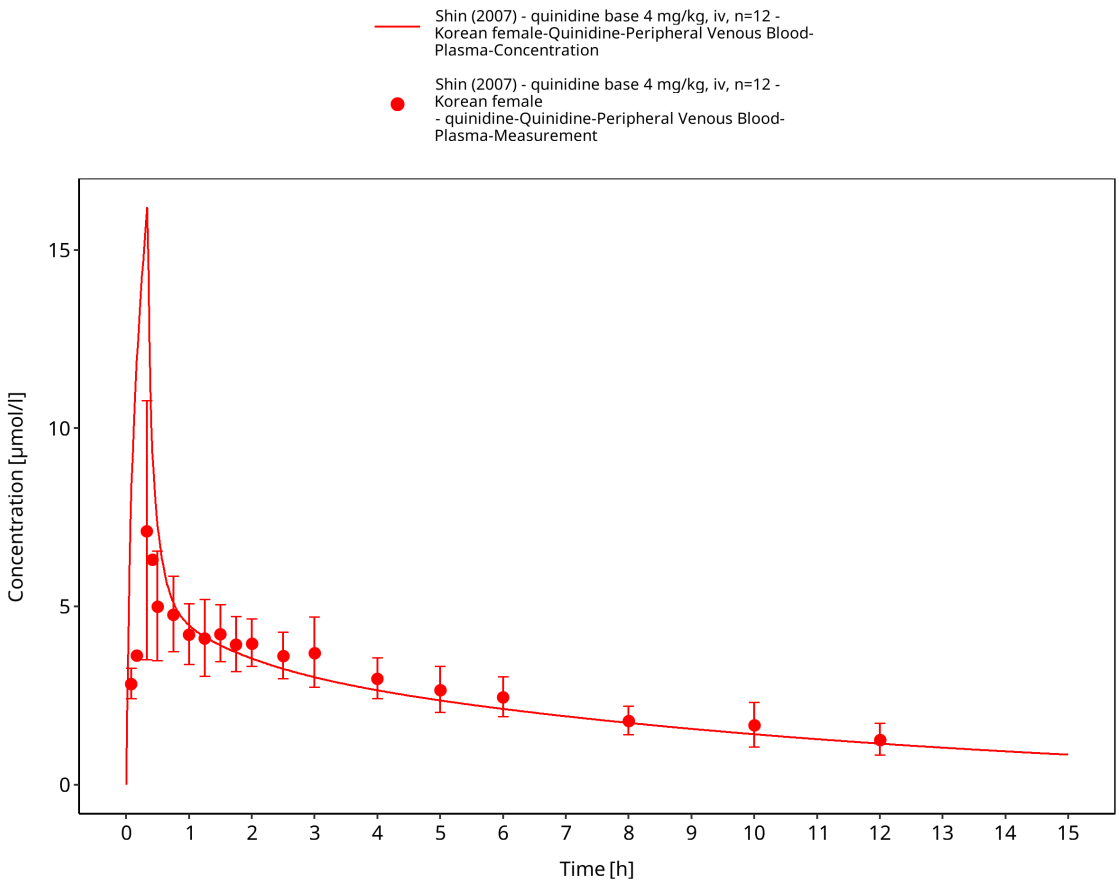


Figure 3-42: Shin 2007: quinidine base 4 mg/kg, iv, n=12 - Korean female

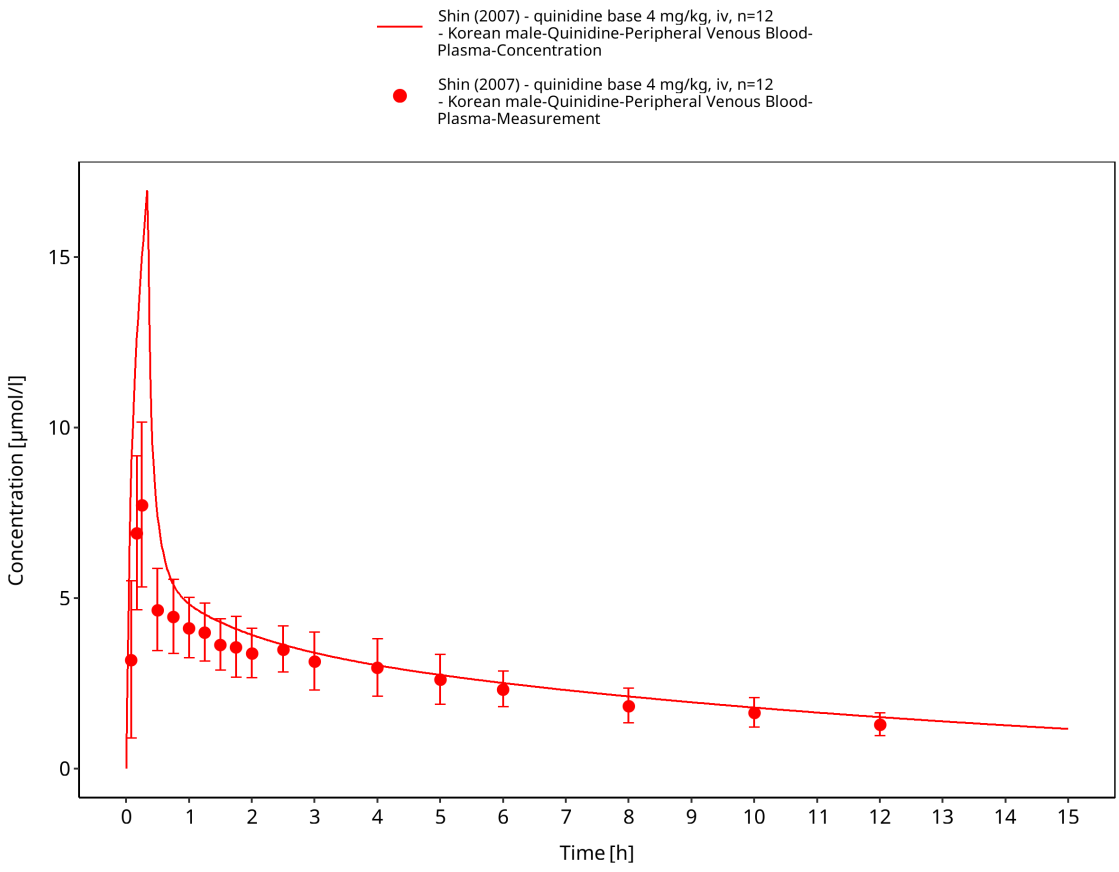


Figure 3-43: Shin 2007: quinidine base 4 mg/kg, iv, n=12 - Korean male

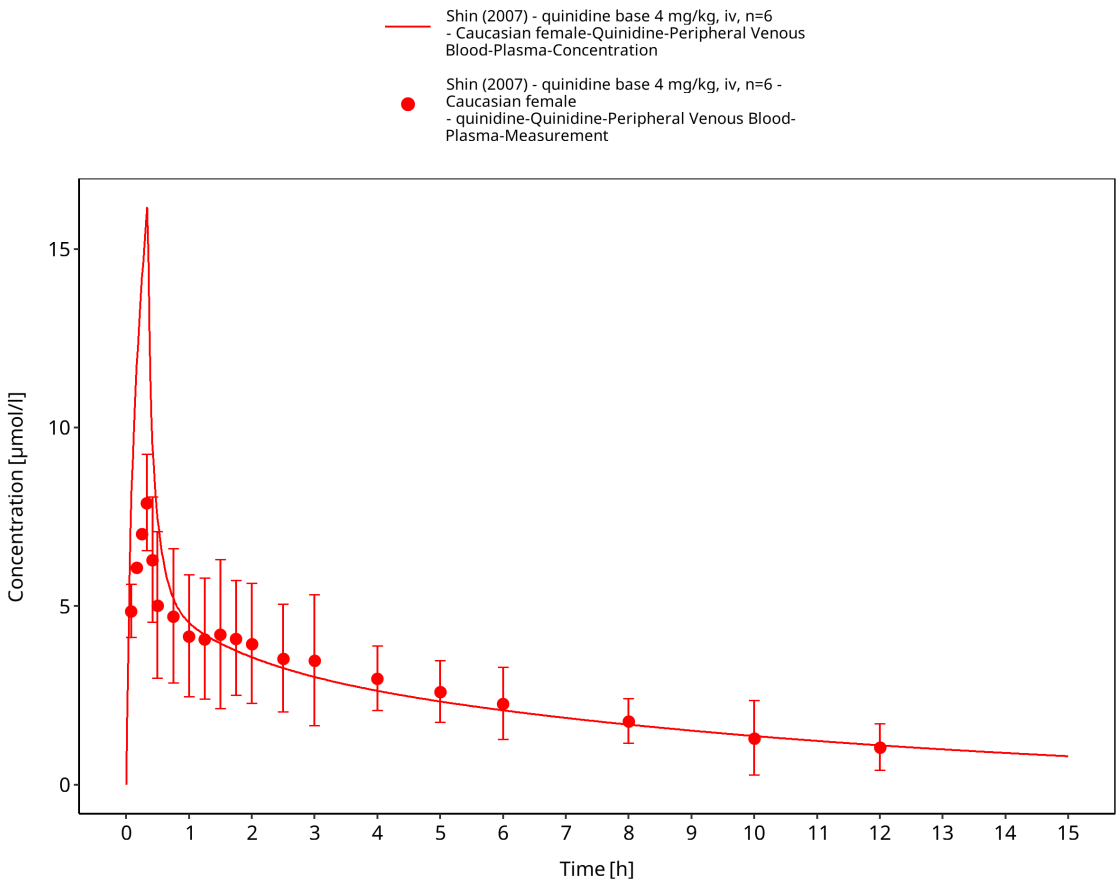


Figure 3-44: Shin 2007: quinidine base 4 mg/kg, iv, n=6 - Caucasian female

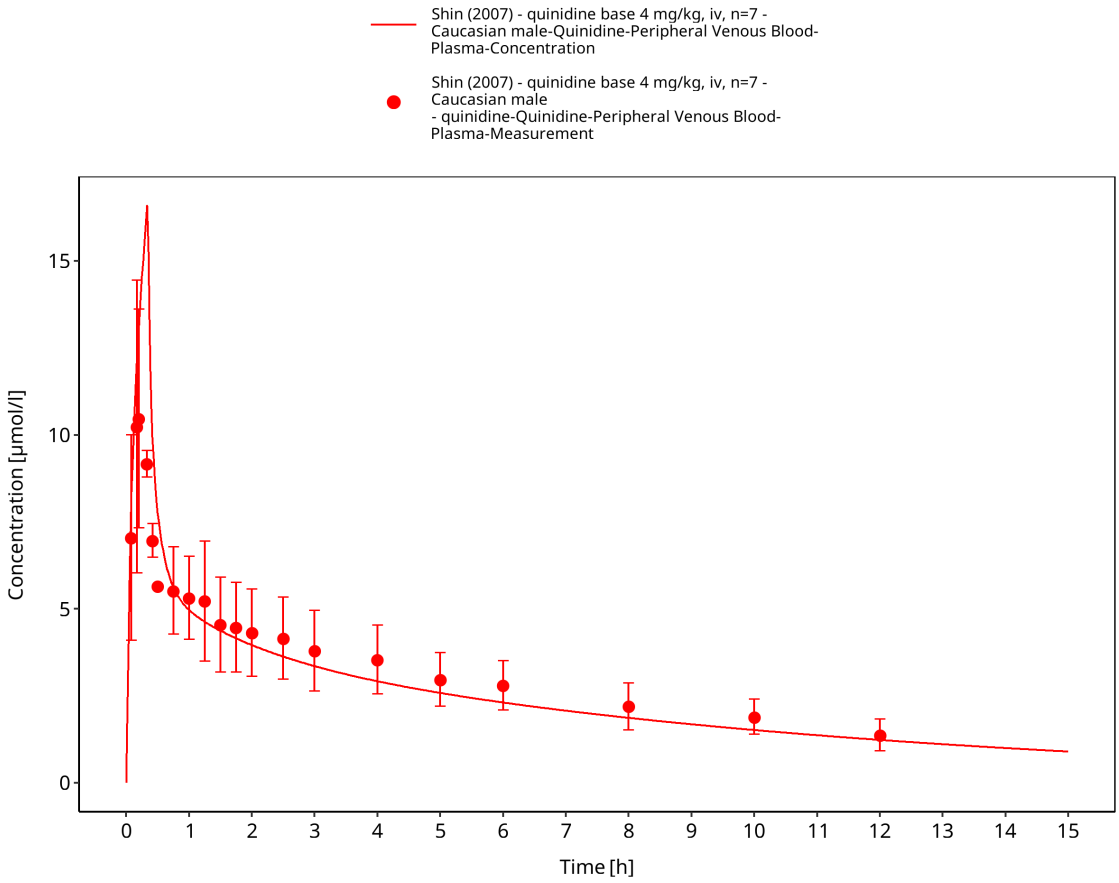


Figure 3-45: Shin 2007: quinidine base 4 mg/kg, iv, n=7 - Caucasian male

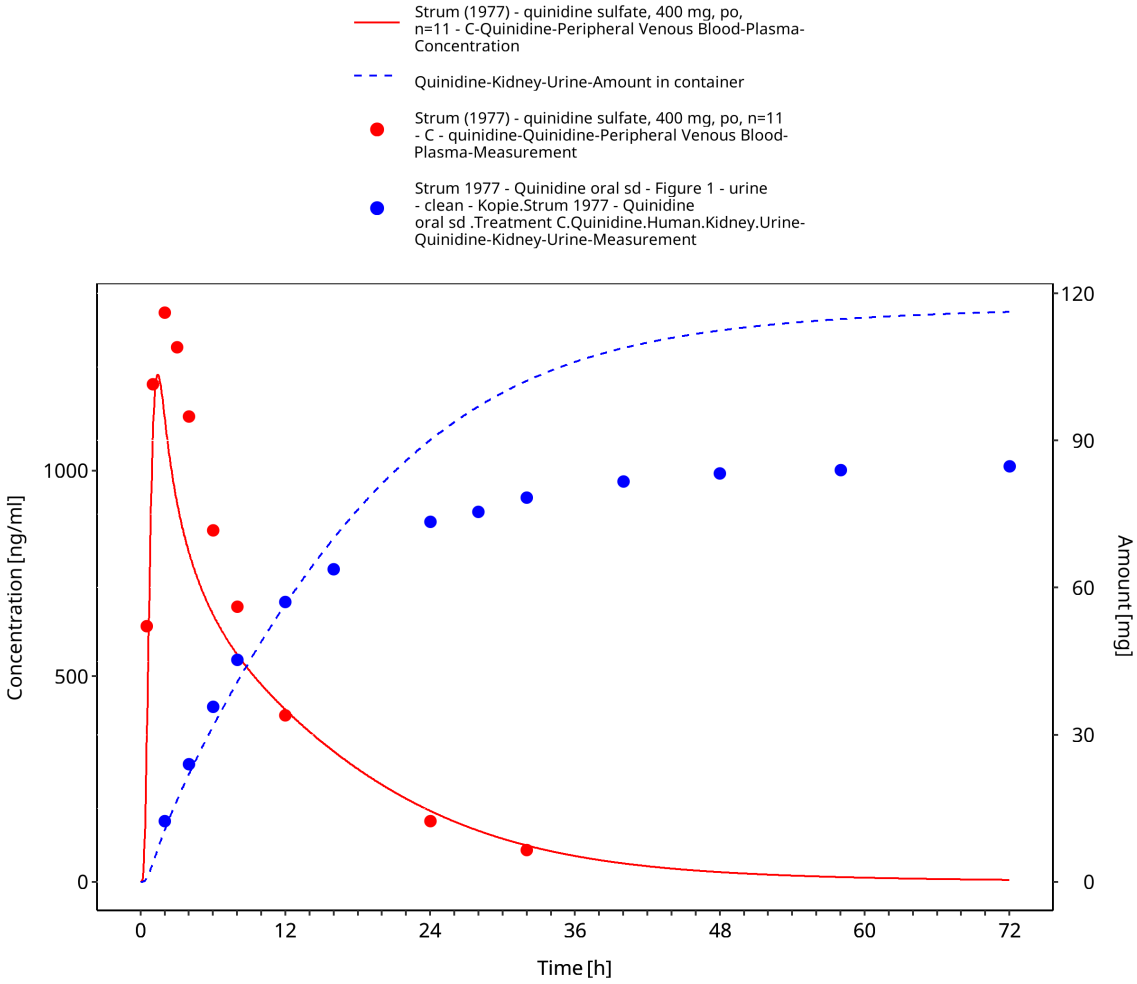


Figure 3-46: Strum 1977: quinidine sulfate, 400 mg, po, n=11 - C

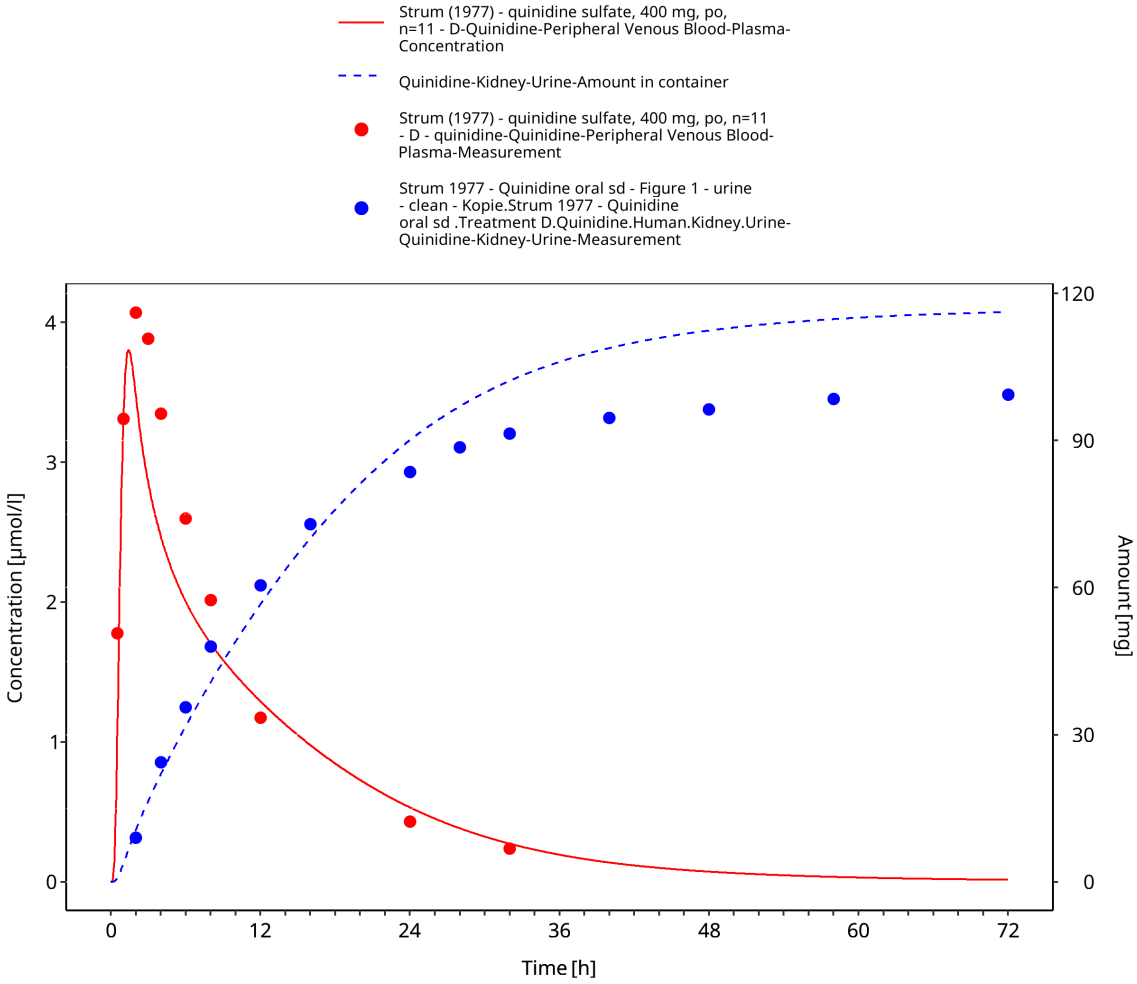


Figure 3-47: Strum 1977: quinidine sulfate, 400 mg, po, n=11 - D

4 Conclusion

The quinidine parent-metabolite PBPK model describes the evaluated plasma concentration-time data for quinidine and 3-hydroxyquinidine after intravenous and oral administration in adults. The evaluation includes single-dose and multiple-dose regimens, solution, capsule, and tablet formulations, total and unbound quinidine observations, and quinidine-only control arms from interaction studies.

The model combines CYP3A4-mediated 3-hydroxyquinidine formation with P-gp transport, passive renal filtration, active renal secretion, residual elimination, enterohepatic circulation, and CYP2D6 inhibition.

The concentration-time profiles and goodness-of-fit diagnostics characterize model performance within the evaluated dose, route, regimen, formulation, analyte, and adult population ranges. No formal acceptance criterion was applied.

The main limitations are sparse 3-hydroxyquinidine and unbound quinidine data, single-subject intravenous profiles, and heterogeneous oral formulations and regimens.

5 References

[1] Feick D, Rüdesheim S, Marok FZ, Selzer D, Loer HLH, Teutonico D, et al. Physiologically-Based Pharmacokinetic Modeling of Quinidine to Establish a CYP3A4, P-gp and CYP2D6 Drug-Drug-Gene Interaction Network. CPT Pharmacometrics Syst Pharmacol. 2023;12:1143-1156. doi: 10.1002/psp4.12981.

[2] Rüdesheim S, Loer HLH, Feick D, Marok FZ, Fuhr LM, Selzer D, et al. A Comprehensive CYP2D6 Drug-Drug-Gene Interaction Network for Application in Precision Dosing and Drug Development. Clin Pharmacol Ther. 2025;117:1718-1731. doi: 10.1002/cpt.3604.

[3] Ochs HR, Grube E, Greenblatt DJ, Woo E, Bodem G. Intravenous quinidine: pharmacokinetic properties and effects on left ventricular performance in humans. Am Heart J. 1980;99:468-475. doi: 10.1016/0002-8703(80)90381-6.

[4] Maeda K, Takano J, Ikeda Y, Fujita T, Oyama Y, Nozawa K, Kumagai Y, Sugiyama Y. Nonlinear pharmacokinetics of oral quinidine and verapamil in healthy subjects: a clinical microdosing study. Clin Pharmacol Ther. 2011;90:263-270. doi: 10.1038/clpt.2011.108.

[5] Andreasen AH, Brøsen K, Damkier P. A comparative pharmacokinetic study in healthy volunteers of the effect of carbamazepine and oxcarbazepine on CYP3A4. Epilepsia. 2007;48:490-496. doi: 10.1111/j.1528-1167.2007.00924.x.

[6] Ochs HR, Greenblatt DJ, Woo E, Franke K, Pfeifer HJ, Smith TW. Single and multiple dose pharmacokinetics of oral quinidine sulfate and gluconate. Am J Cardiol. 1978;41:770-777. doi: 10.1016/0002-9149(78)90830-5.

[7] Strum JD, Colaizzi JL, Jaffe JM, Martineau PC, Poust RI. Comparative bioavailability of four commercial quinidine sulfate tablets. J Pharm Sci. 1977;66:539-542. doi: 10.1002/jps.2600660420.

[8] Ching MS, Elliott SL, Stead CK, Murdoch RT, Devenish-Meares S, Morgan DJ, Smallwood RA. Quinidine single dose pharmacokinetics and pharmacodynamics are unaltered by omeprazole. Aliment Pharmacol Ther. 1991;5:523-531. doi: 10.1111/j.1365-2036.1991.tb00521.x.

[9] Laganière S, Davies RF, Carignan G, et al. Pharmacokinetic and pharmacodynamic interactions between diltiazem and quinidine. Clin Pharmacol Ther. 1996;60:255-264. doi: 10.1016/S0009-9236(96)90052-1.

[10] Bleske BE, Carver PL, Annesley TM, Bleske JRM, Morady F. The effect of ciprofloxacin on the pharmacokinetic and ECG parameters of quinidine. J Clin Pharmacol. 1990;30:911-915. doi: 10.1002/j.1552-4604.1990.tb03570.x.

[11] Damkier P, Hansen LL, Brøsen K. Rifampicin treatment greatly increases the apparent oral clearance of quinidine. Pharmacol Toxicol. 1999;85:257-262. doi: 10.1111/j.1600-0773.1999.tb02019.x.

[12] Edwards DJ, Lavoie R, Beckman H, Blevins R, Rubenfire M. The effect of coadministration of verapamil on the pharmacokinetics and metabolism of quinidine. Clin Pharmacol Ther. 1987;41:68-73. doi: 10.1038/clpt.1987.11.

[13] Hardy BG, Schentag JJ. Lack of effect of cimetidine on the metabolism of quinidine: effect on renal clearance. Int J Clin Pharmacol Ther Toxicol. 1988;26:388-391.

[14] Kolb KW, Garnett WR, Small RE, Vetrovec GW, Kline BJ, Fox T. Effect of cimetidine on quinidine clearance. Ther Drug Monit. 1984;6:306-312. doi: 10.1097/00007691-198409000-00009.

[15] Darbar D, Dell'Orto S, Mörike K, Wilkinson GR, Roden DM. Dietary salt increases first-pass elimination of oral quinidine. Clin Pharmacol Ther. 1997;61:292-300. doi: 10.1016/S0009-9236(97)90161-2.

[16] Fremstad D, Nilsen OG, Storstein L, Amlie J, Jacobsen S. Pharmacokinetics of quinidine related to plasma protein binding in man. Eur J Clin Pharmacol. 1979;15:187-192. doi: 10.1007/BF00563104.

[17] Guentert TW, Holford NH, Coates PE, Upton RA, Riegelman S. Quinidine pharmacokinetics in man: choice of a disposition model and absolute bioavailability studies. J Pharmacokinet Biopharm. 1979;7:315-330. doi: 10.1007/BF01062532.

[18] Mason WD, Covinsky JO, Valentine JL, Kelly KL, Weddle OH, Martz BL. Comparative plasma concentrations of quinidine following administration of one intramuscular and three oral formulations to 13 human subjects. J Pharm Sci. 1976;65:1325-1329. doi: 10.1002/jps.2600650916.

[19] Bolme P, Otto U. Dose-dependence of the pharmacokinetics of quinidine. Eur J Clin Pharmacol. 1977;12:73-76. doi: 10.1007/BF00561409.

[20] Frigo GM, Perucca E, Teggia-Droghi M, Gatti G, Mussini A, Salerno J. Comparison of quinidine plasma concentration curves following oral administration of some short- and long-acting formulations. Br J Clin Pharmacol. 1977;4:449-454. doi: 10.1111/j.1365-2125.1977.tb00760.x.

[21] Wishart DS, Knox C, Guo AC, et al. DrugBank: a comprehensive resource for in silico drug discovery and exploration. Nucleic Acids Res. 2006;34:D668-D672. doi: 10.1093/nar/gkj067.

[22] DrugBank. Quinidine. 2022. DB00908.

[23] Takano J, Maeda K, Bolger MB, Sugiyama Y. The prediction of the relative importance of CYP3A/P-glycoprotein to the nonlinear intestinal absorption of drugs by advanced compartmental absorption and transit model. Drug Metab Dispos. 2016;44:1808-1818. doi: 10.1124/dmd.116.070011.

[24] Tachibana T, Kitamura S, Kato M, et al. Model analysis of the concentration-dependent permeability of P-gp substrates. Pharm Res. 2010;27:442-446. doi: 10.1007/s11095-009-0026-9.

[25] Nielsen TL, Rasmussen BB, Flinois JP, Beaune P, Brøsen K. In vitro metabolism of quinidine: the (3S)-3-hydroxylation of quinidine is a specific marker reaction for cytochrome P-4503A4 activity in human liver microsomes. J Pharmacol Exp Ther. 1999;289:31-37. doi: 10.1016/S0022-3565(24)38103-0.

[26] Moghadamnia AA, Rostami-Hodjegan A, Abdul-Manap R, Wright CE, Morice AH, Tucker GT. Physiologically based modelling of inhibition of metabolism and assessment of the relative potency of drug and metabolite: dextromethorphan vs. dextrorphan using quinidine inhibition. Br J Clin Pharmacol. 2003;56:57-67. doi: 10.1046/j.1365-2125.2003.01853.x.

[27] Austin RP, Barton P, Cockroft SL, Wenlock MC, Riley RJ. The influence of nonspecific microsomal binding on apparent intrinsic clearance, and its prediction from physicochemical properties. Drug Metab Dispos. 2002;30:1497-1503. doi: 10.1124/dmd.30.12.1497.

[28] Watanabe R, Esaki T, Kawashima H, et al. Predicting Fraction Unbound in Human Plasma from Chemical Structure: Improved Accuracy in the Low Value Ranges. Mol Pharm. 2018;15:5302-5311.

doi: 10.1021/acs.molpharmaceut.8b00785.

[29] Rao BR, Rambhau D. Absence of a pharmacokinetic interaction between quinidine and diazepam. Drug Metabol Drug Interact. 1995;12:45-51. doi: 10.1515/dmdi.1995.12.1.45.

[30] Kaukonen KM, Olkkola KT, Neuvonen PJ. Itraconazole increases plasma concentrations of quinidine. Clin Pharmacol Ther. 1997;62:510-517. doi: 10.1016/S0009-9236(97)90046-1.

[31] Shin JG, Kang WK, Shon JH, Arefayene M, Yoon YR, Kim KA, Kim DI, Kim DS, Cho KH, Woosley RL, Flockhart DA. Possible interethnic differences in quinidine-induced QT prolongation between healthy Caucasian and Korean subjects. Br J Clin Pharmacol. 2007;63:206-215. doi: 10.1111/j.1365-2125.2006.02793.x.

[32] Kuepfer L, Niederalt C, Wendl T, Schlender JF, Willmann S, Lippert J, Block M, Eissing T, Teutonico D. Applied Concepts in PBPK Modeling: How to Build a PBPK/PD Model. CPT Pharmacometrics Syst Pharmacol. 2016;5:516-531. doi: 10.1002/psp4.12134.

[33] Willmann S, Höhn K, Edginton A, Sevestre M, Solodenko J, Weiss W, Lippert J, Schmitt W. Development of a physiology-based whole-body population model for assessing the influence of individual variability on the pharmacokinetics of drugs. J Pharmacokinet Pharmacodyn. 2007;34:401-431. doi: 10.1007/s10928-007-9053-5.

[34] Damkier P, Hansen LL, Brøsen K. Effect of fluvoxamine on the pharmacokinetics of quinidine. Eur J Clin Pharmacol. 1999;55:451-456. doi: 10.1007/s002280050655.

[35] Lumen AA, Acharya P, Polli JW, Ayrton A, Ellens H, Bentz J. If the K_I is defined by the free energy of binding to P-glycoprotein, which kinetic parameters define the IC_{50} for the Madin-Darby canine kidney II cell line overexpressing human multidrug resistance 1 confluent cell monolayer? Drug Metab Dispos. 2010;38:260-269. doi: 10.1124/dmd.109.029843.

[36] Ching MS, Blake CL, Ghabrial H, Ellis SW, Lennard MS, Tucker GT, Smallwood RA. Potent inhibition of yeast-expressed CYP2D6 by dihydroquinidine, quinidine, and its metabolites. Biochem Pharmacol. 1995;50:833-837. doi: 10.1016/0006-2952(95)00207-g.

[37] Grube S, Langguth P, Junginger HE, Kopp S, Midha KK, Shah VP, Stavchansky S, Dressman JB, Barends DM. Biowaiver monographs for immediate release solid oral dosage forms: quinidine sulfate. J Pharm Sci. 2009;98:2238-2251. doi: 10.1002/jps.21606.

[38] DrugBank. 3-Hydroxyquinidine. 2022. DBMET00382.

[39] ChemAxon. Chemicalize. 2020. https://chemicalize.com.

[40] Berezhkovskiy LM. Volume of distribution at steady state for a linear pharmacokinetic system with peripheral elimination. J Pharm Sci. 2004;93:1628-1640. doi: 10.1002/jps.20073.

[41] Nguyen CA, Konan-Kouakou YN, Allémann E, Doelker E, Quintanar-Guerrero D, Fessi H, Gurny R. Preparation of surfactant-free nanoparticles of methacrylic acid copolymers used for film coating. AAPS PharmSciTech. 2006;7:E54-E60. doi: 10.1208/pt070363.

[42] Zhang Y, Huo M, Zhou J, Zou A, Li W, Yao C, Xie S. DDSolver: an add-in program for modeling and comparison of drug dissolution profiles. AAPS J. 2010;12:263-271. doi: 10.1208/s12248-010-9185-1.